



# OPEN Agentic AI-driven autonomous decision support system for smart agriculture

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Crop productivity is heavily impacted by inefficient fertilizer usage, improper fertilizer handling, and inappropriately chosen crops. To address these issues, this research work proposes an AI-powered Smart Agriculture Prediction System utilizing intelligent agents for carrying out soil classification, estimating soil parameters, crop suggestion, and fertilizer suggestion. The soil classifier module is trained with 1,563 images of black soils, red soils, clay soils, and alluvial soils using MobileNet-V2, ResNet, and Custom CNN, with Custom CNN resulting in a higher accuracy of 92.88%, which performed better in classifying soils based on textures. A soil parameter estimation agent utilizes regression models for estimating pH and NPK content of soils using images. For crop suggestion, a crop dataset with 2,200 samples with parameters such as N, P, K, T, H, pH, and rainfall is used, in which Random Forest model performed better with an accuracy of 92.4% when compared with CNN and DNN models. For fertilizer suggestion, XGBoost performed better with an accuracy of 94.7% in estimating fertilizers such as Urea, DAP, NPK, Potash, and Compost. Real-time climatic parameters are obtained using API in order to make dynamic updates for climatic parameters. Real-time weather data obtained through APIs enables dynamic updates of climatic parameters, while Explainable AI techniques such as SHAP and LIME enhance model transparency and user trust. Additionally, the system incorporates an interactive agent-based framework that processes user inputs, including location, soil images, and nutrient levels, to generate adaptive outputs such as weather alerts, yield potential, and personalized recommendations. The experimental results demonstrate that the proposed system effectively integrates deep learning, ensemble learning, and explainability to deliver a scalable, efficient, and sustainable decision-support solution for precision agriculture, promoting optimized resource utilization and environmental stewardship.

**Keywords** Soil analysis, Crop prediction, Fertilizer recommendation, Sustainable farming, Machine learning, Accuracy, Data-driven agriculture, Environmental monitoring, Yield optimization

Economic stability is directly reliant on agriculture millions of individuals earn their livelihood from it. And yet farmers continue to struggle to get a reliable grip on soil content, choose the best crops, and determine the most appropriate fertilizers to improve their yields. When they are unable to pinpoint soil nutrients or consider regional environmental conditions, crops falter: yields are reduced, soil quality reduces, and resources are wasted. We need to solve these issues with more intelligent, technology-based solutions if we're going to have sustainable farming and consistent productivity in the long run. Traditional soil analysis methods include physicochemical laboratory tests (e.g., pH measurement, texture analysis using the hydrometer method, and nutrient quantification such as NPK testing) and manual field-based classification techniques (e.g., USDA soil taxonomy or FAO soil texture charts). While these methods are accurate, they are time-consuming, labor-intensive, and require specialized equipment or expert knowledge. Additionally, they are not scalable for rapid, real-time decision-making in modern precision agriculture. With the emergence of Artificial Intelligence (AI) and Machine Learning (ML), new opportunities have opened up to transform agricultural decision-making. The adoption of AI agents in agriculture makes it possible to automate and target precision at different stages — from soil categorization and nutrient estimation to fertilizer and crop forecasting — with better accuracy and responsiveness to real-time environmental conditions. The system utilizes AI-based models to scan soil images, weather conditions, and chemical characteristics to make smart suggestions for crop and fertilizer management.

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The system starts by processing soil images to automatically detect black, red, clay, and alluvial soils using deep learning algorithms. Machine learning algorithms, such as Convolutional Neural Networks and lightweight deep learning models such as MobileNet, have the capability to learn distinguishable features from textures, colors, and other structures in images that are relevant for automatic identification of soils. Unlike traditional machine learning approaches where human interaction is involved in selecting distinguishable features, this technique automatically determines these features from images. Recently, research has found that deep learning algorithms have proven a feasible and efficient alternative approach in classifying soil images in a quicker and more accurate manner compared to conventional soil analysis methods. Through such ML algorithms for classifying soils, the system gains a firm basis in making decisions concerning agricultural production, such as crop and fertilizer recommendations.

The primary contributions of the work proposed are listed below:

- Combining various AI agents for computer-aided soil classification, crop suggestion, and fertilizer prediction.
- Use of weather API data to dynamically change environmental conditions, improving real-time decision-making accuracy.
- Comparative analysis of deep models and ensemble models and showcasing MobileNetV2, Random Forest, and XGBoost as the most effective performers.
- Integration of explainable AI (XAI) methods like SHAP and LIME to ensure transparency in reasons for prediction.
- Design of a single AI-based agricultural pipeline that reduces human intervention and optimizes yield efficiency.

The system employs an interactive, agent-based architecture with user inputs location, soil photos, and nutrient values—directly linked to smart backend processing units. This is a modular design that can ensure scalability and allows for seamless integration of future farming extensions such as pest forecasting and irrigation planning. The front-end developed interface offers results in the form of yield estimation, crop recommendation, fertilizer suggestions, and chatbot-based advisories, making the user experience intuitive and responsive. Through the integration of explainable AI, real-time weather intelligence, and adaptive user interactivity, this framework provides a solid basis for next-generation smart agriculture platforms emphasizing efficiency, sustainability, and empowering farmers.

## Literature survey

Recent advancements in agricultural technology have demonstrated the transformative potential of Artificial Intelligence (AI) and Machine Learning (ML) in improving decision-making across various aspects of farming, including soil classification, crop selection, and fertilizer optimization. Tripathi et al.<sup>1</sup> highlighted the wide range of ML applications in agriculture, emphasizing their role in optimizing crop yield and resource allocation. Akpenpuun et al.<sup>2</sup> reviewed multiple temperature management techniques in greenhouses, revealing that data-driven models significantly enhance microclimatic stability and crop productivity. Similarly, a study on Moroccan greenhouse systems<sup>3</sup> demonstrated that ML algorithms can effectively predict thermal behavior, leading to improved climate control and reduced energy consumption. Modeling studies by<sup>4</sup> and Guillén-Navarro et al.<sup>5</sup> established the accuracy of ML models in forecasting soil temperature and environmental parameters under diverse climatic conditions. Mampitiya et al.<sup>6</sup> further developed an AI-based soil temperature prediction model, achieving high reliability for agricultural planning. In the context of crop recommendation, Rani et al.<sup>7</sup> introduced optimization-based ML models that efficiently mapped soil nutrients and environmental factors to suitable crop types. Monjezi et al.<sup>8</sup> developed hybrid intelligent models integrating ML and fuzzy systems for predicting greenhouse air temperature, achieving superior performance in dynamic conditions. Sharma et al.<sup>9</sup> proposed an advanced framework combining cloud, satellite, and AI-based data for smart farming, which improved accuracy in soil and crop monitoring. Mohan et al.<sup>10</sup> presented an explainable AI (XAI)-enabled approach that enhanced precision crop yield prediction through the integration of AI and interpretable learning frameworks. Dey et al.<sup>11</sup> proposed a machine learning-based crop recommendation framework for Indian agricultural conditions using soil nutrients (NPK), soil pH, and climatic variables, demonstrating the effectiveness of ensemble and tree-based models in improving crop suitability predictions.

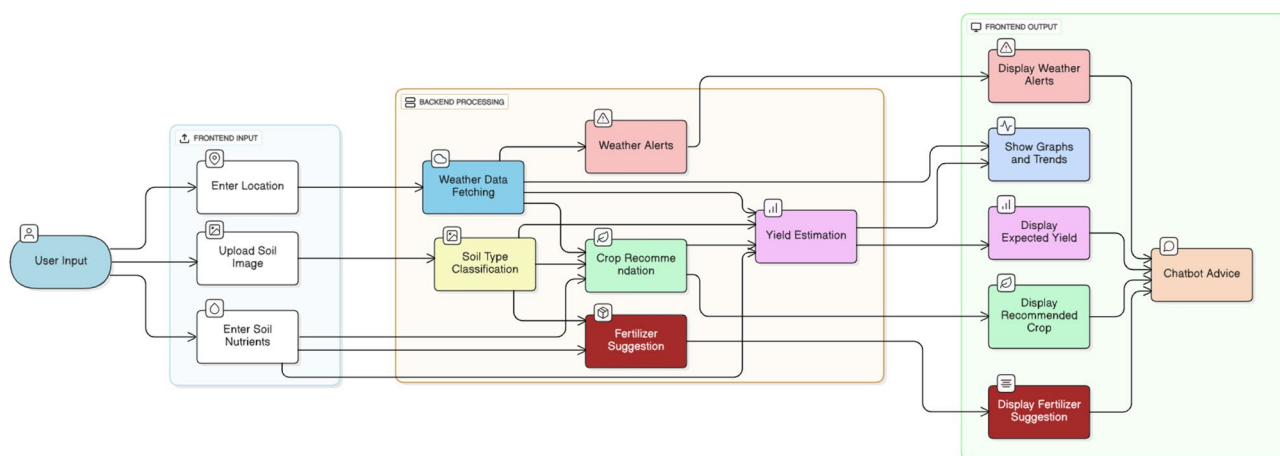
Generative AI has also been explored in agriculture, as highlighted by Reddy et al.<sup>12</sup>, who developed a data-driven precision farming system integrating machine learning techniques with soil and fertilizer data to improve crop and fertilizer recommendation accuracy under real-world farming conditions, and Suganthi et al.<sup>14</sup>, where AI-powered optimization systems and GIS-based deep learning models were utilized to enhance yield prediction and crop mapping. Liu<sup>13</sup> discussed the growing importance of AI in sustainable farming, emphasizing its role in automation and precision management. Sai et al.<sup>15</sup> emphasized the role of Generative AI in Agriculture 4.0, introducing advanced predictive frameworks for smart farming. Crop<sup>16</sup> presented an ensemble machine learning-assisted framework for sustainable food development, demonstrating improved crop and fertilizer recommendation performance through model aggregation strategies. Nandom et al.<sup>17</sup> applied Random Forest and hierarchical clustering techniques for crop and fertilizer recommendation, showing that ensemble-based approaches can effectively support farmer-centric decision-making using soil nutrient data. Musanase et al.<sup>18</sup> developed a data-driven crop and fertilizer recommendation system that revolutionized traditional farming through ML integration. Tanaka et al.<sup>19</sup> evaluated the reliability of ML models in providing fertilizer recommendations, confirming that ensemble models such as XGBoost yielded superior accuracy in diverse conditions. Akhilesh et al.<sup>20</sup> and Khan et al.<sup>21</sup> further demonstrated that context-aware IoT-assisted ML systems enhance fertilizer prediction and reduce environmental impact.

Ndjuluwa, L. N. P et al.<sup>22</sup> presented an Internet of Things (IoT)–assisted, context-aware fertilizer recommendation system that integrates real-time soil and environmental data with machine learning techniques to support precise and adaptive fertilizer decision-making. Pawar et al.<sup>23</sup> proposed a hybrid AI framework combining federated ML/DL models for soil classification, achieving robust generalization across different soil types. Godara and Toshniwal<sup>24</sup> utilized deep learning for analyzing farmers' helpline queries, demonstrating high accuracy in trend prediction. Folorunso et al.<sup>25</sup> introduced GeaGrow, a mobile-based decision support tool using artificial neural networks for soil nutrient prediction and fertilizer optimization, while Pallavi et al.<sup>26</sup> developed a machine learning–based fertilizer recommendation system aimed at improving sustainable crop yield through data-driven nutrient analysis. Panchbhay et al.<sup>27</sup> proposed a modified MobileNet with leaky ReLU and LSTM, incorporating a data balancing technique to improve soil type classification accuracy under imbalanced conditions. Lanjewar and Gurav<sup>28</sup> demonstrated that CNN-based models effectively classify soil images by automatically learning discriminative texture and color features, enabling reliable soil characterization for precision agriculture. Sharma et al.<sup>29</sup> presented AgriCloud, a cloud-based framework for scalable crop monitoring and intelligent decision support, highlighting the advantages of integrating multiple AI modules for real-time, data-driven agricultural recommendations. Collectively, these studies underscore the increasing integration of ML and AI across various agricultural domains—ranging from soil and temperature modeling to fertilizer and crop recommendation. The consistent high accuracy of models such as Random Forest, XGBoost, and CNN demonstrates their potential in building adaptive, data-driven frameworks for sustainable and efficient farming practices.

## Methodology

Soil efficiency would typically be constrained by incorrect soil analysis, incorrect crop choice, and inefficient fertilizer utilization, resulting in low yields and eventual soil degradation. In an attempt to outsmart these constraints, a gigantic AI-Based Agriculture Prediction Framework was created that integrated various clever agents to treat every farming activity systematically. The system comprises three main agents, the Soil Classification Agent, Crop Recommendation Agent, and the Fertilizer Prediction Agent, trained on data extremely curated using machine learning and deep learning models for accuracy and consistency in prediction. The whole architecture and interaction among these agents are depicted in Fig. 1, Architectural Diagram of the Multi-Agent System for Smart Agriculture Prediction, which is the sequential process from identification of soil through estimation of nutrients, fertilizer and crop suggestion. The process begins from the soil image capture, where the system identifies the soil as black, red, clay, or alluvial with the help of a trained convolutional network. Once categorized, the Soil Parameter Agent calculates soil nutrient values—Nitrogen (N), Phosphorus (P), and Potassium (K)—and the pH value of the soil through regression analysis. Computed values and real-time weather information (rainfall, temperature, humidity) dynamically collected from a weather API are assumed as input parameters for fertilizer and crop prediction models.

The Crop Recommendation Agent predicts the most suitable crop based on soil and environmental requirements so that the crop is suitable for the nutrient level and environmental conditions of the area. Finally, the Fertilizer Recommendation Agent suggests best fertilizer mixtures based on predicted crop and soil nutrient deficiencies. In the interest of interpretability and transparency of the prediction, Explainable AI (XAI) models such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) are included in the system. They enable the visualization and quantification of feature contributions, providing open transparency on which individual input features like nitrogen, pH, or temperature are responsible for producing the resultant prediction created by the models. This integration of interpretability guarantees that model decisions are understandable and coherent to enable agronomists and farmers to cross-check AI-inferred insights. The combination of explainable learning models with intelligent agents leads to a holistic, transparent, and contextual system to enhance sustainable agriculture practice and general efficiency of yield.



**Fig. 1.** Architectural Diagram of the Multi-Agent System for Smart Agriculture Prediction.

The system starts with the frontend input module, which serves as an interface for interaction with the intelligent agricultural system. In this step, the user is required to input information such as the geographical location of the farmland and upload an image of the soil. The other information needed is for the nutrients present in the soil. Such information basically defines the physical and chemical properties of farmland. In the backend processing module, real-time and forecasted data pertaining to the weather is obtained via the location provided by the user. They interface with external sources to obtain information such as temperature, rainfall, humidity, and extreme weather. They pre-process this information, which is used within different modules in the system. Correct information with regards to the weather will allow for decisions to be time and region-specific.

1. **Soil Type Classification** - The soil type identification module checks the uploaded image of the soil and nutrient content input parameters. Techniques such as image processing are used to identify visual characteristics like textures and colors in the image, while nutrient content parameters allow analysis based on chemistry. Based on these identified parameters, machine learning algorithms classify the type of soil present. Classification of soil type is necessary to identify suitable crops and fertilizer methods.
2. **Crop Recommendation** - With regards to the classified type of soil used in planting, depending on the retrieved weather information, the crop selection module recommends which crop is ideal to plant. This module employs models that have been trained based on the relationship between soils, climatic elements, and crop performance in the past.
3. **Fertilizer suggestion** - The fertilizer suggestion module makes recommendations based on nutrient deficiencies in the soil and nutrients necessary for the recommended crop. Through the integration of soil type results and crop information, the system makes recommendations on which fertilizer to apply. This enables effective nutrient application, prevents overapplication of fertilizer, and promotes sustainable crop production practices.
4. **Yield Estimation** - The Yield Estimation module predicts crop yield based on soil properties, climatic factors, and recommended crops to be grown. Computer algorithms called predictive models, using historical agricultural data, forecast inputs before an actual crop can be grown. Such forecasting informs farmers on resource allocation, financial management, and expectations concerning crop output.
5. **Weather Alerts** - The weather alert module is responsible for consistently assessing the projected weather conditions in an attempt to detect any potential dangers such as a drought, floods, heat, or a storm. In light of such unfavorable conditions, notifications are sent to the output interface.
6. **Weather Data Fetching** - The module utilizes the OpenWeather API to obtain real-time and forecasted weather information based on the user's geographic location. It collects key parameters such as temperature, rainfall, humidity, and weather conditions. This data enables accurate crop recommendations, yield prediction, and timely weather alerts.

The Frontend Output Module serves as the visualization and interaction layer of the Soil2Harvest-AI framework. It presents processed insights in a clear, intuitive, and farmer-centric format to support informed decision-making. This module displays real-time weather alerts, interactive graphs illustrating climatic and soil trends, predicted crop yield, recommended crops, and optimized fertilizer suggestions. By translating complex backend analytics into actionable insights, the module ensures accessibility for users with varying levels of technical expertise. The integration of an intelligent chatbot further enhances usability by providing contextual explanations, personalized guidance, and real-time assistance, thereby improving user engagement and decision confidence.

CropiBot is the conversational AI assistant embedded within the Soil2Harvest-AI platform, designed to deliver personalized agricultural advisory services. Acting as an intelligent agent, CropiBot interacts with users through natural language, enabling them to query system outputs, seek clarifications, and receive actionable recommendations. CropiBot leverages backend outputs such as soil classification results, weather forecasts, crop recommendations, fertilizer suggestions, and yield predictions to generate context-aware responses. It assists farmers in understanding why a particular crop or fertilizer is recommended, explains weather alerts and their potential impact, and provides best-practice guidance for cultivation and resource management. By offering continuous, real-time support, CropiBot bridges the gap between advanced AI-driven analytics and practical field-level decision-making. This conversational interface not only improves system transparency but also empowers farmers to make data-driven decisions confidently, promoting sustainable and optimized agricultural practices.

### Technology and setup

The system was developed using Python 3.10 on Google Colab GPU for efficient model training and inference. Machine learning and deep learning libraries such as TensorFlow, Keras, and Scikit-learn were used for building deep learning and ensemble models, while Pandas and NumPy supported data manipulation and numerical computations. Visual interpretability and model explainability were implemented using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), which identify the contribution of individual features to the model's predictions. The architecture is modular, integrating multiple AI agents into a centralized pipeline that manages data flow from soil image input to prediction output. The front-end interface, developed with Streamlit, allows farmers to upload soil images or input nutrient values and receive real-time crop and fertilizer suggestions. Additionally, a live weather API dynamically updates environmental parameters such as temperature, humidity, and rainfall to ensure recommendations remain consistent with prevailing climatic conditions.

## Datasets used

Three separate datasets were utilized for this study, each with its own specific role in the Smart Agriculture Prediction System: soil classification, crop suggestion, and fertilizer suggestion. Soil Classification Dataset consisted of a total of 1,563 labeled images of soil, systematically arranged into four major categories of soil — black, red, clay, and alluvial — that account for the majority of the major agricultural regions' dominant soil classes. Each photo captured the natural surface color and textural variation typical of its soil class, offering crucial visual information for classification. The data was gathered from Kaggle and public-domain agricultural repositories, where all images were already labeled according to soil type, ensuring consistency in annotation and variability in lighting conditions, moisture levels, and geographic distribution. It added associated metadata like geographic position, soil moisture category, and texture density index, which helped the model learn about spatial and textural variation between soil types. The Crop Recommendation Dataset contained 2,200 structured instances, each noting seven crucial agronomic factors determining the crop yield and adaptability — nitrogen (N), phosphorus (P), and potassium (K) levels as reflecting the composition of soil nutrients; temperature and humidity as indicating atmospheric and climatic conditions; pH value to reflect soil acidity or alkalinity; and rainfall to determine the hydrological contribution to crop growth. The data set also included corresponding labels for the best crop to be grown in such conditions, ranging from classes like rice, maize, wheat, sugarcane, cotton, and pulses. Each entry was a balanced blend of environmental as opposed to soil factors, with well-represented coverage over a variety of climatic zones and levels of soil fertility. The Fertilizer Recommendation Dataset contained approximately 1,200 samples, one for each set of variations in soil type, crop type, and corresponding fertilizer class. Features were soil nutrient percentages (N, P, K), type of soil, type of crop, and matching fertilizer label from the six broad fertilizer classes — Urea, DAP, MOP, NPK, Potash, and Compost. They consisted of organic and inorganic fertilizers, allowing the model to learn a full mapping of the nutrient imbalance to the corrective fertilizer treatment. The data reflected real-world relationships employed in current agronomy, mapping nutrient deficiency to related fertilizer needs for different crop-soil combinations. Each sample reflected realistic agronomic conditions gathered via a mixture of regional agricultural data and expert-curated synthetic supplementation to provide data balance between crop varieties and soil types. Together, these three datasets offered a multi-level basis for the system, wherein the soil classification module provided critical inputs for the prediction of crop suitability and the ensuing chosen crop then informed fertilizer recommendation choices — creating a solid, data-driven pipeline for precision-based agricultural insight generation.

The agricultural data sources, traits, and usage described in the Table 1 titled Agricultural Data Sources, Features, and Intended Use Cases above make up a broad and diverse platform for making well-informed

| Dataset                           | Sample data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Features                                                                       | Purpose                                        |           |            |          |                        |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
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| Soil Image Dataset                | <div><div>Red Soil</div><div>Alluvial Soil</div><div>Black Soil</div><div>Clay Soil</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Color (RGB, HSV), Texture (GLCM, LBP), Moisture indicators, Soil type patterns | Soil type classification, soil health analysis |           |            |          |                        |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| Crop Recommendation Dataset       | <table><thead><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th></tr><tr><th>N</th><th>P</th><th>K</th><th>temperatu</th><th>humidity</th><th>pH</th><th>rainfall</th><th>label</th><th></th></tr></thead><tbody><tr><td>90</td><td>42</td><td>43</td><td>20.87974</td><td>82.00274</td><td>6.502985</td><td>202.9355</td><td>rice</td><td></td></tr><tr><td>85</td><td>58</td><td>41</td><td>21.77046</td><td>80.31964</td><td>7.038096</td><td>226.6555</td><td>rice</td><td></td></tr><tr><td>60</td><td>55</td><td>44</td><td>23.00446</td><td>82.32076</td><td>7.840207</td><td>263.9642</td><td>rice</td><td></td></tr><tr><td>74</td><td>35</td><td>40</td><td>26.4911</td><td>80.15836</td><td>6.980401</td><td>242.864</td><td>rice</td><td></td></tr><tr><td>78</td><td>42</td><td>42</td><td>20.13017</td><td>81.60487</td><td>7.628473</td><td>262.7173</td><td>rice</td><td></td></tr><tr><td>69</td><td>37</td><td>42</td><td>23.05805</td><td>83.37012</td><td>7.073454</td><td>251.055</td><td>rice</td><td></td></tr><tr><td>69</td><td>55</td><td>38</td><td>22.70884</td><td>82.63941</td><td>5.700806</td><td>271.3249</td><td>rice</td><td></td></tr><tr><td>94</td><td>53</td><td>40</td><td>20.27774</td><td>82.89409</td><td>5.718627</td><td>241.9742</td><td>rice</td><td></td></tr><tr><td>89</td><td>54</td><td>38</td><td>24.51588</td><td>83.53522</td><td>6.685346</td><td>230.4462</td><td>rice</td><td></td></tr><tr><td>68</td><td>58</td><td>38</td><td>23.22397</td><td>83.03323</td><td>6.336254</td><td>221.2092</td><td>rice</td><td></td></tr><tr><td>91</td><td>53</td><td>40</td><td>26.52724</td><td>81.41754</td><td>5.386168</td><td>264.6149</td><td>rice</td><td></td></tr><tr><td>90</td><td>46</td><td>42</td><td>23.97898</td><td>81.45062</td><td>7.502834</td><td>250.0832</td><td>rice</td><td></td></tr><tr><td>78</td><td>58</td><td>44</td><td>26.8008</td><td>80.88685</td><td>5.108682</td><td>284.4365</td><td>rice</td><td></td></tr><tr><td>93</td><td>56</td><td>36</td><td>24.01498</td><td>82.05687</td><td>6.984354</td><td>185.2773</td><td>rice</td><td></td></tr><tr><td>94</td><td>50</td><td>37</td><td>25.66585</td><td>80.66385</td><td>6.94802</td><td>209.587</td><td>rice</td><td></td></tr><tr><td>60</td><td>48</td><td>39</td><td>24.28209</td><td>80.30026</td><td>7.042299</td><td>231.0863</td><td>rice</td><td></td></tr></tbody></table>                                                                                                                                                                                              |                                                                                | A                                              | B         | C          | D        | E                      | F | G | H | N | P | K | temperatu | humidity    | pH        | rainfall   | label |                        | 90 | 42 | 43 | 20.87974 | 82.00274 | 6.502985 | 202.9355 | rice |      | 85   | 58 | 41 | 21.77046 | 80.31964 | 7.038096 | 226.6555 | rice |      | 60   | 55   | 44 | 23.00446 | 82.32076 | 7.840207 | 263.9642 | rice  |      | 74   | 35   | 40   | 26.4911 | 80.15836 | 6.980401 | 242.864 | rice |     | 78   | 42   | 42   | 20.13017 | 81.60487 | 7.628473 | 262.7173 | rice |    | 69    | 37   | 42   | 23.05805 | 83.37012 | 7.073454 | 251.055 | rice |    | 69 | 55    | 38   | 22.70884 | 82.63941 | 5.700806 | 271.3249 | rice |    | 94 | 53 | 40    | 20.27774 | 82.89409 | 5.718627 | 241.9742 | rice |  | 89 | 54 | 38 | 24.51588 | 83.53522 | 6.685346 | 230.4462 | rice |  | 68 | 58 | 38 | 23.22397 | 83.03323 | 6.336254 | 221.2092 | rice |      | 91 | 53 | 40 | 26.52724 | 81.41754 | 5.386168 | 264.6149 | rice |      | 90   | 46 | 42 | 23.97898 | 81.45062 | 7.502834 | 250.0832 | rice |      | 78  | 58   | 44 | 26.8008 | 80.88685 | 5.108682 | 284.4365 | rice |      | 93   | 56   | 36   | 24.01498 | 82.05687 | 6.984354 | 185.2773 | rice |      | 94   | 50   | 37   | 25.66585 | 80.66385 | 6.94802 | 209.587 | rice |    | 60  | 48   | 39   | 24.28209 | 80.30026 | 7.042299 | 231.0863 | rice |     | N, P, K, Temperature, Humidity, pH, Rainfall, Label | Recommended best crop for given soil & climate |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
|                                   | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | B                                                                              | C                                              | D         | E          | F        | G                      | H |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| N                                 | P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | K                                                                              | temperatu                                      | humidity  | pH         | rainfall | label                  |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 90                                | 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 43                                                                             | 20.87974                                       | 82.00274  | 6.502985   | 202.9355 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 85                                | 58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 41                                                                             | 21.77046                                       | 80.31964  | 7.038096   | 226.6555 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 60                                | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 44                                                                             | 23.00446                                       | 82.32076  | 7.840207   | 263.9642 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 74                                | 35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 40                                                                             | 26.4911                                        | 80.15836  | 6.980401   | 242.864  | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 78                                | 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 42                                                                             | 20.13017                                       | 81.60487  | 7.628473   | 262.7173 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 69                                | 37                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 42                                                                             | 23.05805                                       | 83.37012  | 7.073454   | 251.055  | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 69                                | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 38                                                                             | 22.70884                                       | 82.63941  | 5.700806   | 271.3249 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 94                                | 53                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 40                                                                             | 20.27774                                       | 82.89409  | 5.718627   | 241.9742 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 89                                | 54                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 38                                                                             | 24.51588                                       | 83.53522  | 6.685346   | 230.4462 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 68                                | 58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 38                                                                             | 23.22397                                       | 83.03323  | 6.336254   | 221.2092 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 91                                | 53                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 40                                                                             | 26.52724                                       | 81.41754  | 5.386168   | 264.6149 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 90                                | 46                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 42                                                                             | 23.97898                                       | 81.45062  | 7.502834   | 250.0832 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 78                                | 58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 44                                                                             | 26.8008                                        | 80.88685  | 5.108682   | 284.4365 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 93                                | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 36                                                                             | 24.01498                                       | 82.05687  | 6.984354   | 185.2773 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 94                                | 50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 37                                                                             | 25.66585                                       | 80.66385  | 6.94802    | 209.587  | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 60                                | 48                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 39                                                                             | 24.28209                                       | 80.30026  | 7.042299   | 231.0863 | rice                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| Fertilizer Recommendation Dataset | <table><thead><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th><th>I</th></tr><tr><th>N</th><th>P</th><th>K</th><th>rainfall_mm</th><th>temperatu</th><th>humidity_%</th><th>pH</th><th>recommended_fertilizer</th><th></th><th></th></tr></thead><tbody><tr><td>22</td><td>56</td><td>97</td><td>71.4</td><td>33.4</td><td>67.7</td><td>6.28</td><td>Urea</td><td></td><td></td></tr><tr><td>23</td><td>79</td><td>92</td><td>113.4</td><td>14.3</td><td>72.1</td><td>4.73</td><td>Urea</td><td></td><td></td></tr><tr><td>24</td><td>5</td><td>68</td><td>297.8</td><td>28.5</td><td>68.9</td><td>4.53</td><td>Urea</td><td></td><td></td></tr><tr><td>2</td><td>95</td><td>63</td><td>132</td><td>11.4</td><td>97.9</td><td>5.43</td><td>Urea</td><td></td><td></td></tr><tr><td>8</td><td>11</td><td>25</td><td>146.1</td><td>10.4</td><td>95.4</td><td>6.75</td><td>Urea</td><td></td><td></td></tr><tr><td>34</td><td>57</td><td>6</td><td>211.6</td><td>23.2</td><td>29.8</td><td>6.48</td><td>Urea</td><td></td><td></td></tr><tr><td>28</td><td>40</td><td>54</td><td>205.5</td><td>19.4</td><td>61.6</td><td>6.69</td><td>Urea</td><td></td><td></td></tr><tr><td>22</td><td>38</td><td>78</td><td>270.6</td><td>27.9</td><td>93.7</td><td>4.85</td><td>Urea</td><td></td><td></td></tr><tr><td>28</td><td>89</td><td>84</td><td>229.2</td><td>26.2</td><td>66.9</td><td>8.38</td><td>Urea</td><td></td><td></td></tr><tr><td>10</td><td>49</td><td>69</td><td>24.4</td><td>22.7</td><td>51.6</td><td>5.67</td><td>Urea</td><td></td><td></td></tr><tr><td>27</td><td>12</td><td>39</td><td>241.2</td><td>28.2</td><td>94.1</td><td>7.1</td><td>Urea</td><td></td><td></td></tr><tr><td>39</td><td>16</td><td>38</td><td>46.7</td><td>21.1</td><td>73.5</td><td>7.16</td><td>Urea</td><td></td><td></td></tr><tr><td>13</td><td>69</td><td>103</td><td>53.5</td><td>31.4</td><td>80.9</td><td>6.75</td><td>Urea</td><td></td><td></td></tr><tr><td>31</td><td>31</td><td>13</td><td>219</td><td>13.3</td><td>55.1</td><td>5.31</td><td>Urea</td><td></td><td></td></tr><tr><td>35</td><td>100</td><td>8</td><td>274.1</td><td>17.5</td><td>52.8</td><td>7.52</td><td>Urea</td><td></td><td></td></tr><tr><td>5</td><td>36</td><td>75</td><td>101.1</td><td>14.8</td><td>94.4</td><td>7.73</td><td>Urea</td><td></td><td></td></tr><tr><td>27</td><td>32</td><td>112</td><td>72.2</td><td>36.8</td><td>63.1</td><td>7.73</td><td>Urea</td><td></td><td></td></tr><tr><td>26</td><td>31</td><td>66</td><td>20.1</td><td>20.6</td><td>44.4</td><td>5.16</td><td>Urea</td><td></td><td></td></tr></tbody></table> |                                                                                | A                                              | B         | C          | D        | E                      | F | G | H | I | N | P | K         | rainfall_mm | temperatu | humidity_% | pH    | recommended_fertilizer |    |    | 22 | 56       | 97       | 71.4     | 33.4     | 67.7 | 6.28 | Urea |    |    | 23       | 79       | 92       | 113.4    | 14.3 | 72.1 | 4.73 | Urea |    |          | 24       | 5        | 68       | 297.8 | 28.5 | 68.9 | 4.53 | Urea |         |          | 2        | 95      | 63   | 132 | 11.4 | 97.9 | 5.43 | Urea     |          |          | 8        | 11   | 25 | 146.1 | 10.4 | 95.4 | 6.75     | Urea     |          |         | 34   | 57 | 6  | 211.6 | 23.2 | 29.8     | 6.48     | Urea     |          |      | 28 | 40 | 54 | 205.5 | 19.4     | 61.6     | 6.69     | Urea     |      |  | 22 | 38 | 78 | 270.6    | 27.9     | 93.7     | 4.85     | Urea |  |    | 28 | 89 | 84       | 229.2    | 26.2     | 66.9     | 8.38 | Urea |    |    | 10 | 49       | 69       | 24.4     | 22.7     | 51.6 | 5.67 | Urea |    |    | 27       | 12       | 39       | 241.2    | 28.2 | 94.1 | 7.1 | Urea |    |         | 39       | 16       | 38       | 46.7 | 21.1 | 73.5 | 7.16 | Urea |          |          | 13       | 69       | 103  | 53.5 | 31.4 | 80.9 | 6.75 | Urea     |          |         | 31      | 31   | 13 | 219 | 13.3 | 55.1 | 5.31     | Urea     |          |          | 35   | 100 | 8                                                   | 274.1                                          | 17.5 | 52.8 | 7.52 | Urea |  |  | 5 | 36 | 75 | 101.1 | 14.8 | 94.4 | 7.73 | Urea |  |  | 27 | 32 | 112 | 72.2 | 36.8 | 63.1 | 7.73 | Urea |  |  | 26 | 31 | 66 | 20.1 | 20.6 | 44.4 | 5.16 | Urea |  |  | N, P, K, Rainfall_mm, Temperature, Humidity_%, pH, Recommended Fertilizer | Suggest optimal fertilizer type & quantity |
|                                   | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | B                                                                              | C                                              | D         | E          | F        | G                      | H | I |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| N                                 | P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | K                                                                              | rainfall_mm                                    | temperatu | humidity_% | pH       | recommended_fertilizer |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 22                                | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 97                                                                             | 71.4                                           | 33.4      | 67.7       | 6.28     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 23                                | 79                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 92                                                                             | 113.4                                          | 14.3      | 72.1       | 4.73     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 24                                | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 68                                                                             | 297.8                                          | 28.5      | 68.9       | 4.53     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 2                                 | 95                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 63                                                                             | 132                                            | 11.4      | 97.9       | 5.43     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 8                                 | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 25                                                                             | 146.1                                          | 10.4      | 95.4       | 6.75     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 34                                | 57                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6                                                                              | 211.6                                          | 23.2      | 29.8       | 6.48     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 28                                | 40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 54                                                                             | 205.5                                          | 19.4      | 61.6       | 6.69     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 22                                | 38                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 78                                                                             | 270.6                                          | 27.9      | 93.7       | 4.85     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 28                                | 89                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 84                                                                             | 229.2                                          | 26.2      | 66.9       | 8.38     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 10                                | 49                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 69                                                                             | 24.4                                           | 22.7      | 51.6       | 5.67     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 27                                | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 39                                                                             | 241.2                                          | 28.2      | 94.1       | 7.1      | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 39                                | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 38                                                                             | 46.7                                           | 21.1      | 73.5       | 7.16     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 13                                | 69                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 103                                                                            | 53.5                                           | 31.4      | 80.9       | 6.75     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 31                                | 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 13                                                                             | 219                                            | 13.3      | 55.1       | 5.31     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 35                                | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8                                                                              | 274.1                                          | 17.5      | 52.8       | 7.52     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 5                                 | 36                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 75                                                                             | 101.1                                          | 14.8      | 94.4       | 7.73     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 27                                | 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 112                                                                            | 72.2                                           | 36.8      | 63.1       | 7.73     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |
| 26                                | 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 66                                                                             | 20.1                                           | 20.6      | 44.4       | 5.16     | Urea                   |   |   |   |   |   |   |           |             |           |            |       |                        |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |      |      |    |          |          |          |          |       |      |      |      |      |         |          |          |         |      |     |      |      |      |          |          |          |          |      |    |       |      |      |          |          |          |         |      |    |    |       |      |          |          |          |          |      |    |    |    |       |          |          |          |          |      |  |    |    |    |          |          |          |          |      |  |    |    |    |          |          |          |          |      |      |    |    |    |          |          |          |          |      |      |      |    |    |          |          |          |          |      |      |     |      |    |         |          |          |          |      |      |      |      |      |          |          |          |          |      |      |      |      |      |          |          |         |         |      |    |     |      |      |          |          |          |          |      |     |                                                     |                                                |      |      |      |      |  |  |   |    |    |       |      |      |      |      |  |  |    |    |     |      |      |      |      |      |  |  |    |    |    |      |      |      |      |      |  |  |                                                                           |                                            |

**Table 1.** Agricultural data Sources, Features, and intended use Cases.

decisions in smart agriculture. The Soil Image Dataset describes the soil in terms of color parameters in both RGB and HSV channels, texture information using GLCM and LBP methods, and visual signs of water content. Such parameters help in reliable categorization of soils with a focus on health assessment by recognizing tangible nutrient and composition structures associated with nutrient and water-holding capacity. The Crop Recommendation Dataset brings into consideration major nutrient composition in soils such as Nitrogen, Phosphorus, and Potassium content in addition to climatic parameters such as temperature, humidity, pH level, and rainfall. Such a well-organized and numerical representation of important crop parameters provides credible support in making decisions for crop selection based on an in-depth systematic study related to climatic and soil parameters under which crops grow in an optimized manner. The Fertilizer Recommendation Dataset brings into consideration major nutrient composition in addition to climatic parameters and soil pH with a focus on providing recommendations on the most suitable fertilizer based on type and quantity. Such datasets in combination make up a broad platform in a smart agriculture system facilitating support in assessment related to soils, crop selection, and nutrient requirements in a manner that boosts agricultural sustainability and efficiency in a productive manner.

### Data preprocessing

The preprocessing process was an important step in cleaning the data to maintain the accuracy, consistency, and quality of the data prior to training models for soil classification, crop suggestion, and fertilizer suggestion. For the soil image dataset, all raw images initially underwent a review to eliminate duplicates, blurred images, and unnecessary backgrounds such that only clean and informative samples were kept. Every image was resized to a standard resolution of 224 by 224 pixels to ensure consistency in input sizes and normalized in the range 0 to 1 to stabilize the convolutional neural network learning process. In an effort to enhance model generalization and reduce overfitting, a good variety of data augmentation methods were used, such as random rotations, horizontal and vertical flip, brightness, and zooming. These augmentations assisted in the mimicry of true-world variations like illumination, moisture, and soil texture differences, enhancing the model to be more resistant to various environmental factors. In the case of the crop recommendation dataset, all numerical features ranging from nitrogen, phosphorus, and potassium to temperature, humidity, pH, and rainfall were examined with cautious scrutiny for missing, contradictory, and extreme values. Missing values were imputed through statistical imputation techniques like mean or median filling according to feature distributions, and extreme outliers were identified through the interquartile range method and set to a suitable limit to preserve data integrity. All numerical features were standardized through z-score normalization to make sure that all variables had an equal contribution during model training to avoid features with more extensive ranges like rainfall or temperature dominating the learning process. Correlation analysis was conducted with Pearson's coefficient to detect multicollinearity, and redundant features were dropped to ease computation and enhance interpretation. Categorical labels for crop names were converted into numerical form through label encoding to ensure compatibility with supervised learning machine algorithms. In the fertilizer recommendation dataset, categorical features like soil type and crop type were represented in numeric format through one-hot encoding, whereas continuous features like nutrient levels, pH, temperature, humidity, and rainfall were min-max normalized to reduce all attributes to the range 0 to 1. Class imbalance between fertilizer categories, especially for organic fertilizers like compost, was addressed through the use of the Synthetic Minority Oversampling Technique (SMOTE), where extra synthetic samples of minority classes were produced in the feature space to provide enhanced balance without warping data distribution.

All three data sets were run through validation checks for consistency and reliability. Logical filtering was used to remove biologically unrealistic values, including negative rainfall or nutrient concentration and pH levels beyond the viable agricultural range of 3.5 to 9.5. Descriptive statistics and plots like histograms and correlation heatmaps were employed to validate correctness of data distribution and inter-feature relationships. The preprocessed datasets resulting from the above process were clean, normalized, and balanced, and thus formed a robust platform for fast and consistent training of the AI-driven Smart Agriculture Prediction System.

### Weather data processing pipeline

Figure 2 shows that weather data processing chain begins with retrieving the real-time weather data from an Open Weather API using a local system. The obtained information is then preprocessed in a local setting with a Python script in which it gets cleaned, validated, and formatted. Once preprocessed, the information is transmitted to a VPC setting in a safe manner where it is formatted in a standard JSON format. The final step in this processing chain is uploading this JSON-formatted information into an AWS S3 bucket in this VPC setup.

### Model training

The model training phase forms the core of the proposed AI-driven Agriculture Prediction Framework, where a combination of deep learning and ensemble learning techniques was utilized to build high-accuracy models for The Soil Classification Module incorporated three deep learning architectures: MobileNetV2, ResNet, and a Custom Convolutional Neural Network (CNN). As shown in Fig. 3, MobileNetV2 (Mobile Neural Network Version 2) is a lightweight convolutional model that uses depthwise separable convolutions and inverted residuals with linear bottlenecks, optimizing both accuracy and computational efficiency. It reduces the number of parameters while maintaining rich spatial feature representation, making it ideal for real-time agricultural environments. The model was trained for 100 epochs using the Adam optimizer and ReLU activation function, achieving an accuracy of 89.68%, which demonstrates strong performance among all soil classification models. Each model was trained using an 80:20 ratio for training and testing data to ensure generalization and prevent overfitting. Neural network-based architectures were trained for 100 epochs to ensure convergence and effective feature learning.

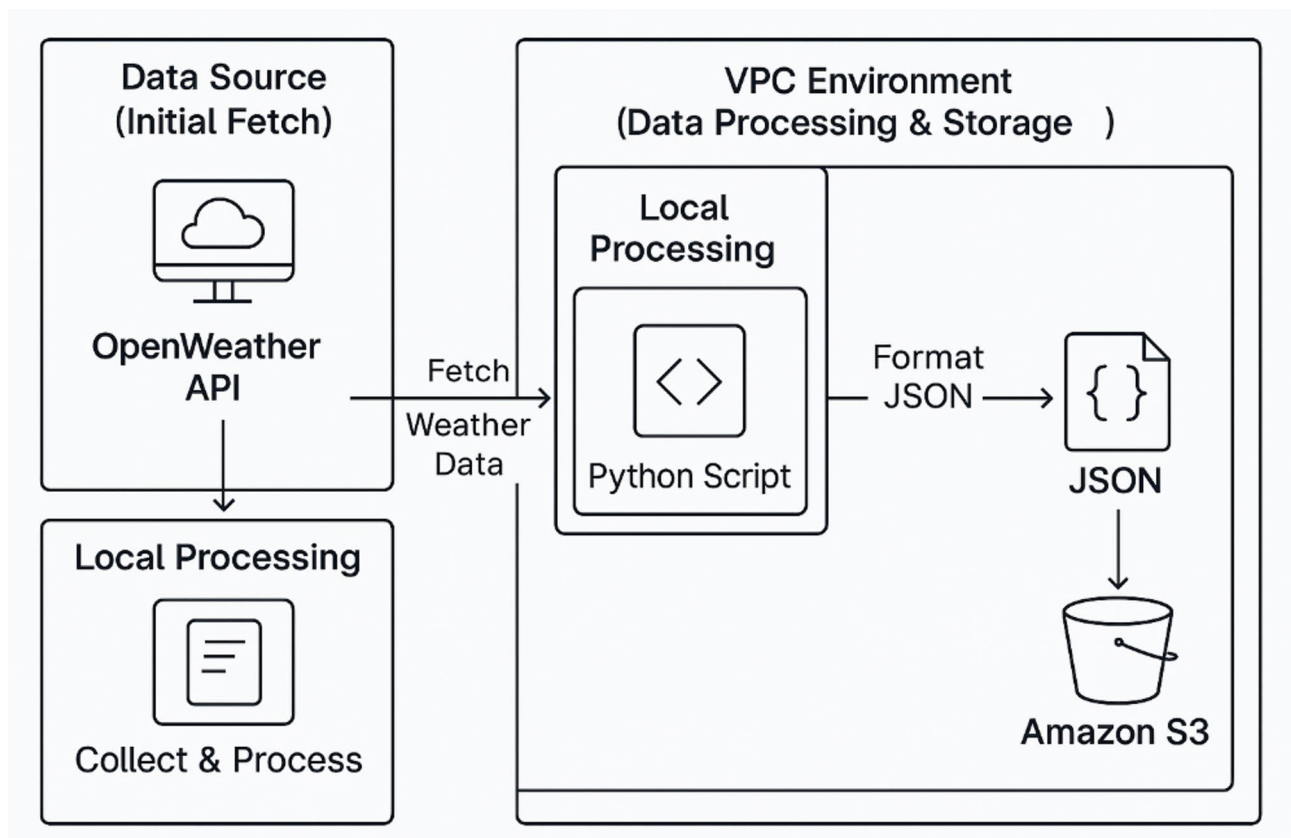


Fig. 2. Block Diagram of Weather Data Acquisition, Local Processing, and Cloud Storage Pipeline.

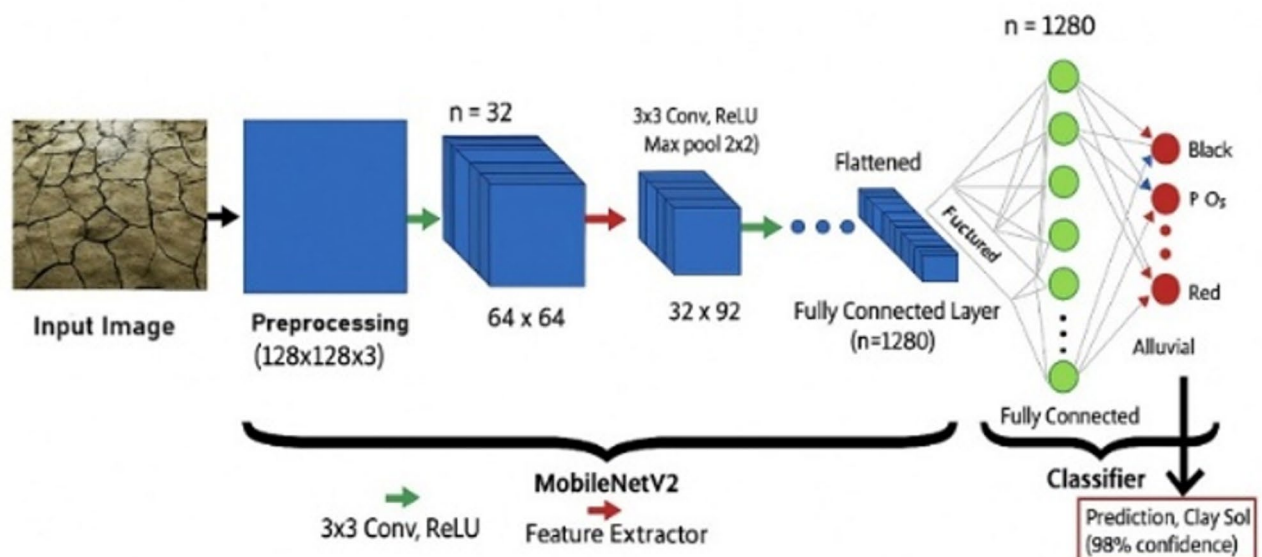
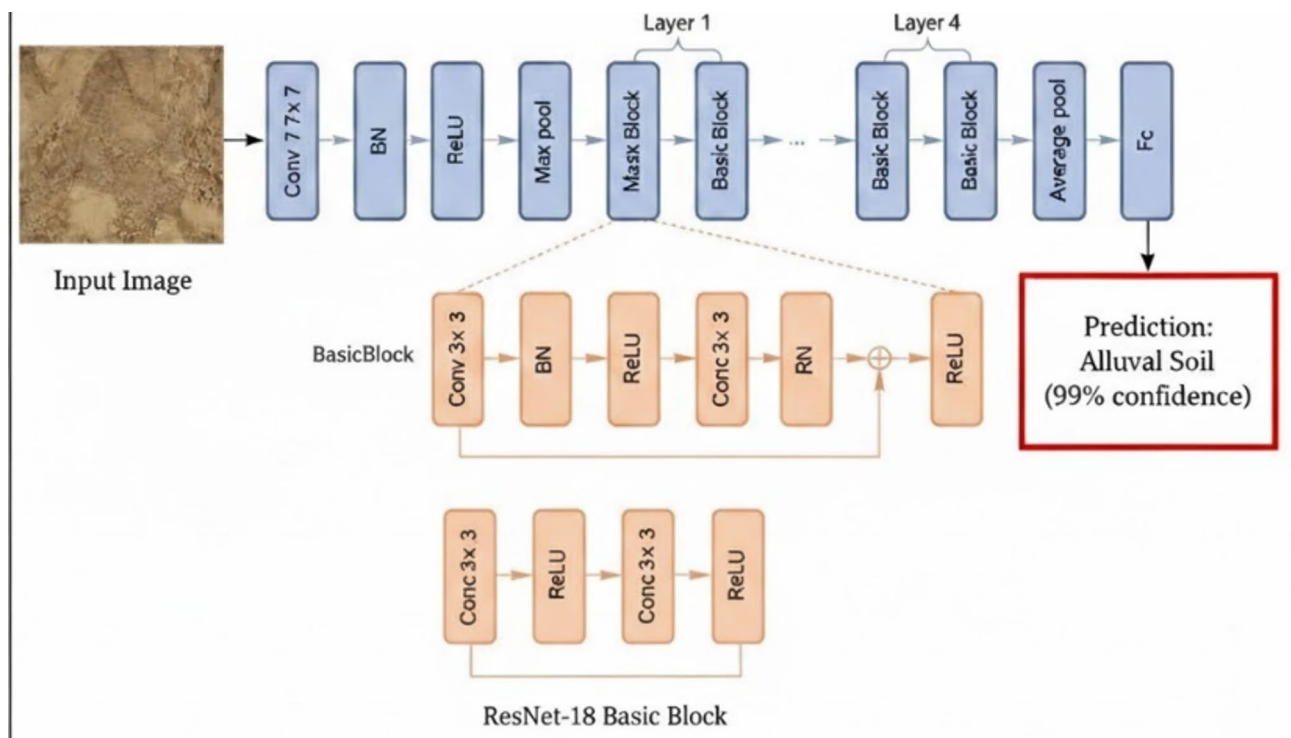
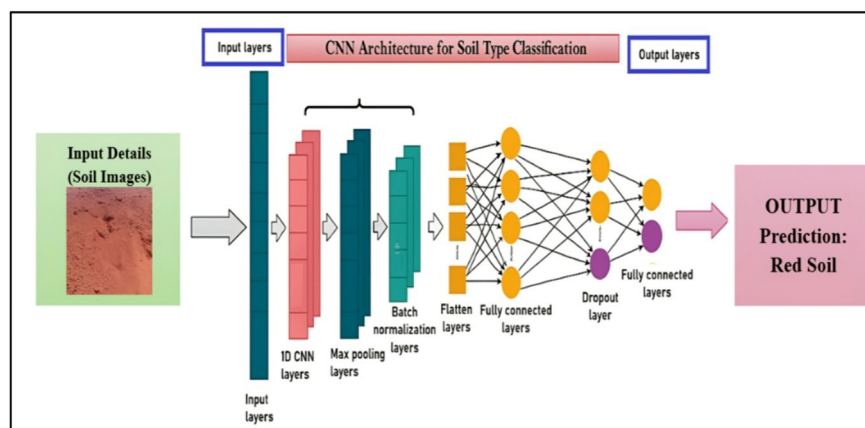


Fig. 3. Architectural Diagram of MobileNetV2 for Soil Image Classification.

The ResNet(Residual Network) structure, depicted in Fig. 4, was created to avoid vanishing gradient problems with the inclusion of residual learning via skip connections. Every residual block is trained on the residual difference between the input and output features, not a direct mapping, enabling deeper networks to learn well without performance loss. ResNet showed great feature reuse, insensitivity to lighting variability, and stability



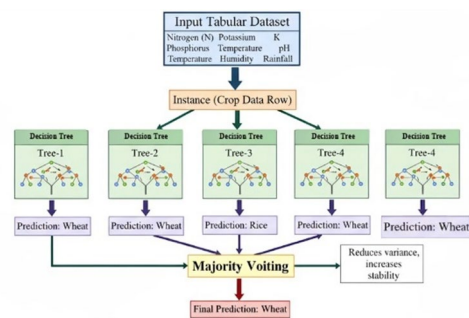
**Fig. 4.** Architectural Diagram of ResNet for Soil Image Classification.



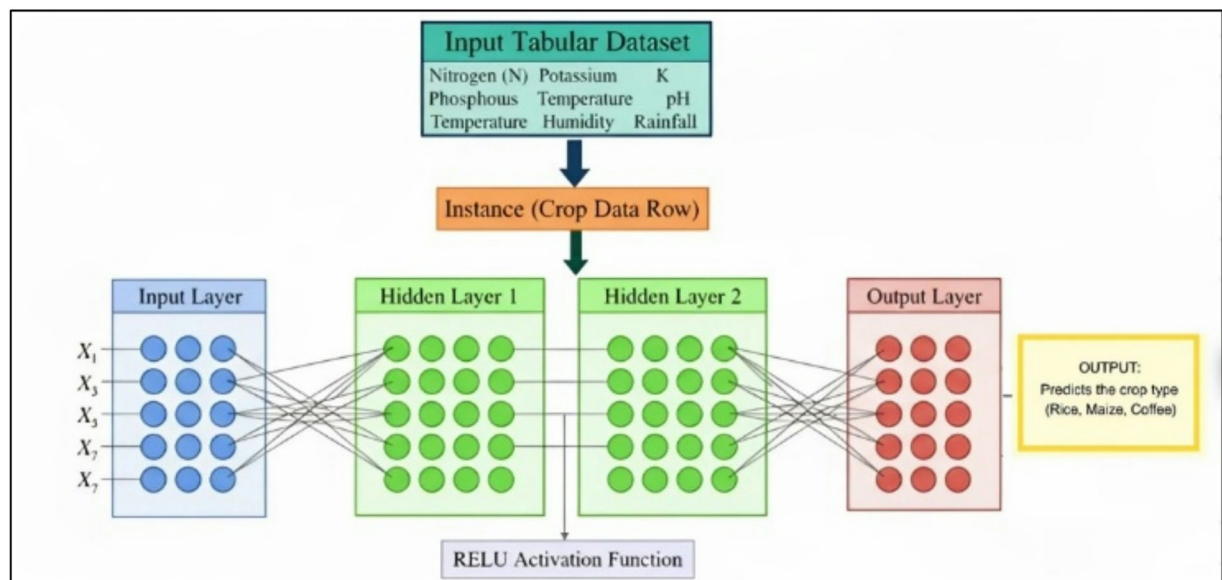
**Fig. 5.** Architectural Diagram of CNN for Soil Image Classification.

in different soil textures, with an accuracy rate of 91.18%. Another Custom CNN, as shown in Fig. 5, was also created to set a baseline structure for soil image classification. It had multiple convolutional and pooling layers to learn hierarchical spatial patterns and then fully connected layers for ultimate classification. ReLU activation function and Softmax output layer were utilized for non-linearity and multi-class prediction, respectively. Though it attained a slightly poorer accuracy of 90.62%, the model yielded useful insights in filter optimization and feature hierarchy that helped in optimizing deeper architectures.

Three machine learning and deep learning models were employed by Crop Recommendation Module: Random Forest, Deep Neural Network (DNN), and Convolutional Neural Network (CNN). Figure 6 makes it clear that Random Forest operates as an ensemble model. It creates multiple decision trees and then pulls their predictions together through majority voting. That method really boosts the stability of predictions while cutting down on variance in a noticeable way. The model processed tabular data that featured elements such as nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall. Random Forest ended up with an accuracy rate of 92.4%. This result positions it as the most reliable option for crop prediction. The Deep Neural Network (DNN) model, as seen in Fig. 7, was used to detect intricate non-linear dependencies among environmental and soil variables. DNN consisted of several hidden layers of densely connected neurons that used ReLU activation



**Fig. 6.** Architectural Diagram of Random Forest for Crop Recommendation.



**Fig. 7.** Architectural Diagram of DNN for Crop Recommendation.

functions, allowing it to pick up complex dependencies between variables. It was trained to 100 epochs and had an accuracy rate of 89%. While being slightly worse performing compared to Random Forest, it performed exceptionally well with continuous-valued inputs and minor variations in the data. The 1D CNN model for crop recommendation, as presented in Fig. 8, made use of 1D convolutional layers for identifying patterns in the features throughout the input space of parameters. In this architecture, convolution and pooling operations were used to filter out noise and obtain significant patterns that are meaningful for crop suitability. The model reached an accuracy rate of 91.8%, verifying the applicability of CNNs even to organized agricultural data.

The Fertilizer Recommendation Module used Random Forest and XGBoost (Extreme Gradient Boosting) classifiers to forecast the best fertilizer type, such as Urea, DAP, MOP, NPK, Potash, and Compost. Figure 9 describes the Random Forest model combined several decision trees to reduce overfitting and worked well on all the combinations of soil and crops. Figure 10 describes how the XGBoost model generalized this strategy by using a gradient boosting framework that incrementally optimized weak learners, both precision and recall. XGBoost obtained a remarkable accuracy of 98.5% with macro-average precision, recall, and F1-score of 94%, outperforming all the other classifiers in the module. In the following stage, hyperparameter tuning was done to tune each model's parameter settings such as learning rate, depth, and batch size for optimal trade-offs between accuracy and computational cost. In the end, we checked out all the models with standard performance metrics. These included things like accuracy, precision, recall, and the F1-score. On top of that, we brought in explainable AI tools such as SHAP, which stands for SHapley Additive exPlanations. We also used LIME, or Local Interpretable Model-agnostic Explanations. The goal was to visualize how important different features were. This setup helped with understanding the models overall. It gave a clear view into why each prediction turned out the way it did.

### Hyperparameter tuning

Tuning hyperparameters plays a key role in adjusting machine learning and deep learning models. It helps get them to work at their best level. In contrast to model parameters, which are learned implicitly during training,

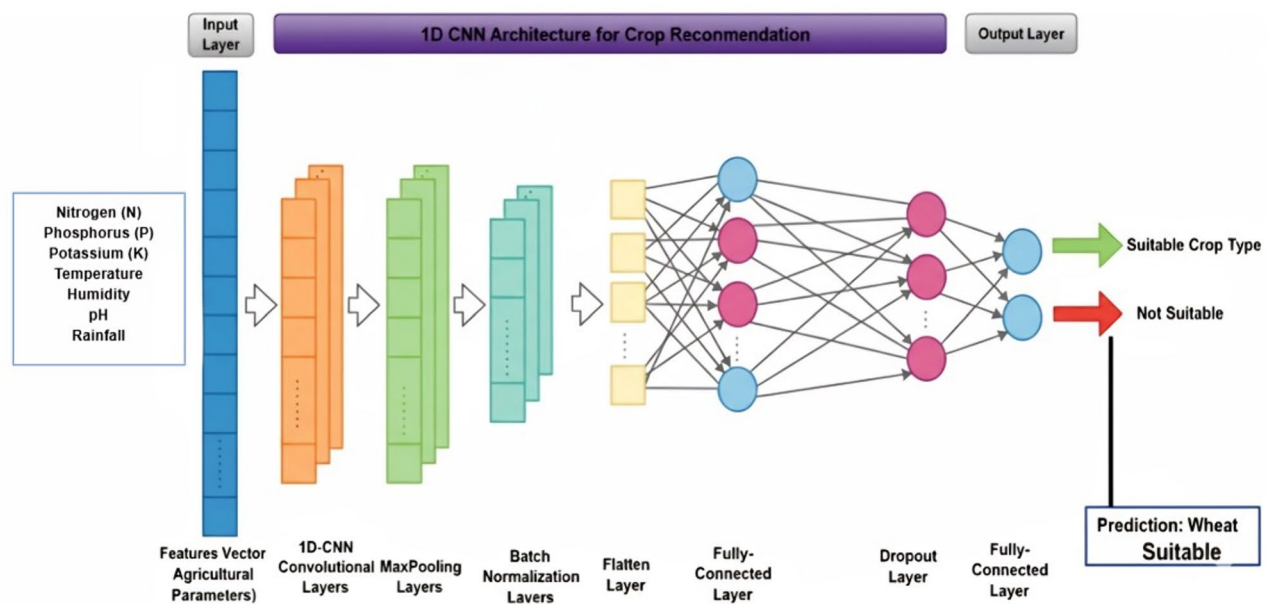


Fig. 8. Architectural Diagram of CNN for Crop Recommendation.

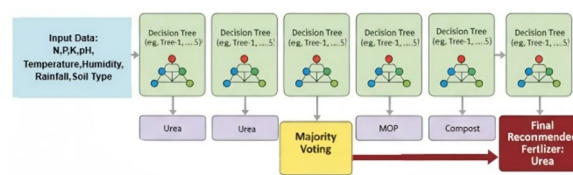


Fig. 9. Architectural Diagram of the Random Forest Model for Fertilizer Recommendation.

hyperparameters are outside settings that regulate the learning process, network architecture, and optimization dynamics. Proper tuning guarantees the model attains a balance that is optimal in terms of preventing underfitting as well as overfitting, enhancing generalizability and predictive precision on unseen data. Systematic hyperparameter tuning was applied in this research to each of the models in soil classification, crop suggestion, and fertilizer prediction modules in order to enhance efficiency as well as consistency in findings. For deep learning models like MobileNetV2, ResNet, and Custom CNN, grid search and iterative fine-tuning methods were used to ascertain the optimal learning rate, type of optimizer, dropout rate, batch size, and number of epochs. The learning rate was also varied between 0.0001 and 0.01, the Adam optimizer being most consistent in convergence throughout the epochs. Batch sizes were tried at 32 and 64, the former giving better updates to the gradient and efficiency in the computations. Dropout regularization values ranging from 0.3 to 0.5 were used to avoid overfitting in networks especially those deeper like ResNet. Early stopping mechanisms were also incorporated to stop training when the validation loss reached a plateau, so the model would still have its best weights without being overtrained. The width multiplier and resolution scaling factors were tweaked in MobileNetV2 to sacrifice accuracy for computation speed, while in the Custom CNN, the number of convolutional filters and kernel sizes were manually optimized to improve spatial feature extraction.

For Deep Neural Network (DNN) and CNN models applied in crop recommendation, the process of tuning involved optimizing the number of hidden layers, number of neurons per layer, and activation functions. ReLU activations were used with better non-linearity management, and batch normalization provided better gradient stability and convergence rate. The best configuration was achieved at three hidden layers of 128, 64, and 32 neurons, respectively, trained for 100 epochs using the learning rate of 0.001. The CNN model kernel size and stride were adjusted for balancing spatial abstraction with training efficiency so that significant feature extraction is realized without over-smoothing important crop data relationships. By contrast, for ensemble models like Random Forest and XGBoost, hyperparameter optimization meant tuning structural parameters governing model complexity and ensemble heterogeneity. They tuned the number of trees in Random Forest from 100 to 300. That parameter is called `n_estimators`. The maximum depth for those trees went from 8 to 20 as well. This setup helped avoid too much over-branching in the model. They also adjusted the `max_features` option. It decides how many features to consider at each split. The goal was to strike a balance between diversity and overall accuracy.

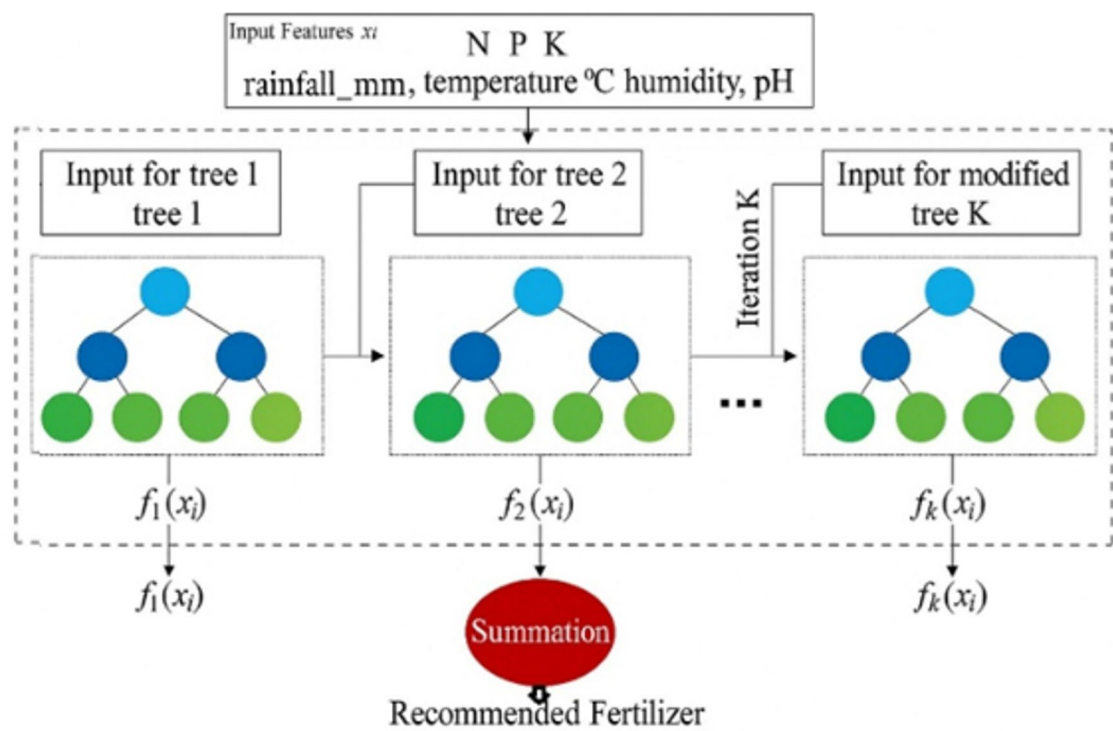


Fig. 10. Architectural Diagram of the XGBoost Model for Fertilizer Recommendation.

| Module                    | Best model    | Hyperparameters                | Best values                       |
|---------------------------|---------------|--------------------------------|-----------------------------------|
| Soil classification       | Custom CNN    | Learning rate                  | 0.001                             |
|                           |               | Optimizer                      | Adam                              |
|                           |               | Batch size                     | 32                                |
|                           |               | Dropout rate                   | 0.3–0.5                           |
|                           |               | Convolution filters            | Manually optimized per layer      |
| Crop recommendation       | Random Forest | n_estimators                   | 100                               |
|                           |               | Max depth                      | 8–20 (optimized in this range)    |
|                           |               | Max features                   | Adjusted per split (as mentioned) |
| Fertilizer recommendation | XGBoost       | Learning rate (eta)            | 0.05                              |
|                           |               | Max depth                      | 8                                 |
|                           |               | Boosting steps                 | 200                               |
|                           |               | Regularization (lambda, alpha) | Tuned to reduce overfitting       |

Table 2. Fine-Tuning parameters for Best-Performing Models.

For XGBoost, the team optimized several things. These included the learning rate, known as eta. They also worked on the number of estimators and maximum depth. Regularization terms like lambda and alpha got attention too. In the end, a learning rate of 0.05 worked best. The depth was set to 8. And they used 200 boosting steps. This combination delivered the highest accuracy and smooth convergence. L1 and L2 regularization played a key role. They effectively cut down on overfitting issues. Hyperparameter tuning made a real difference overall. It boosted the robustness of every model involved. Interpretability improved too. Each agent in the system ran at peak performance. And the whole thing stayed efficient from a computational standpoint. The improved models showed better convergence behavior, improved accuracy, and stable generalization performance across various agricultural conditions. This optimization provided the basis for the next phase of evaluation and analysis, in which the prediction capacities of the models were evaluated using various performance measures. Table 2 summarizes the hyperparameters used for the best-performing models in each module of the Smart Agriculture Prediction System. It highlights the values and ranges considered during fine-tuning, including learning rates, optimizers, batch sizes, network configurations, and ensemble parameters. These settings were selected based on iterative tuning to achieve optimal predictive performance for soil classification, crop recommendation, and fertilizer prediction.

### Evaluation metrics

All the datasets utilized in this study were split with extreme caution into training and testing sets in order to test the performance of the models rigorously. This split has been maintained uniformly across all the datasets, and 80% of the data has been utilized for training and 20% for testing. In this way, one can get an unbiased view of how well the models generalize on unseen data and perform in realistic dental healthcare scenarios. Beyond that, the training set of the Panoramic Teeth Images Dataset was further split into 80% for training and 20% for validation to ensure thorough evaluation and optimization of the models during training. After partitioning the datasets, the models were trained with the training data and tested with the testing data. Performance metrics including accuracy, precision, recall, and F1 score were calculated to gauge the performance of the models<sup>22</sup>.

#### (1) ACCURACY

It is the ratio between correct predictions amongst total instances in the data set. Equation 1 depicts how to calculate accuracy. It gives an overall sense of model performance but is not ideal in the case of imbalanced data sets.

$$\text{Accuracy} = \frac{\text{Number of correct Prediction}}{\text{Total Number of Predictions}} \quad (1)$$

#### (2) PRECISION

Precision measures the actual accuracy of positive predictions from the model. Equation 2 describes how to compute the number of actual positives to be among the set of predicted positives, that is, precision.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

#### (3) RECALL

Recall or sensitivity is the ratio of actual positives that the model will predict correctly. Equation 3 depicts how to compute recall to give an indication of the sensitivity of the model in detecting instances that are actually positive.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3)$$

#### (4) F1 SCORE

F1 score is the harmonic mean of precision and recall. It gives one number that balances both measures. Equation 3 explains the calculation of F1-score. This will be handy if you ever want to seek a balance between precision and recall

$$F1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

These metrics are crucial for evaluating the performance of machine learning models in tasks such as dental caries detection. They provide insights into how well the models classify instances and are essential for comparing different models and selecting the most effective ones for deployment in real-world applications.

### Baseline implementation and fair comparison protocol

To ensure methodological fairness in baseline comparisons, all competing models were trained and evaluated under identical experimental conditions. This includes consistent data preprocessing, identical feature sets, uniform train–test splits, and the same evaluation metrics across all models.

For deep learning–based baselines, identical data augmentation, normalization procedures, batch sizes, and optimization strategies were applied. Classical machine learning baselines were trained using the same preprocessed inputs and evaluated using the same cross-validation or hold-out protocols.

Furthermore, hyperparameter tuning for all models—including baseline methods—was performed using the same search strategy (grid/random search) within comparable parameter ranges. Each model was trained until convergence or early stopping using identical stopping criteria. This ensures that the reported performance improvements reflect genuine model capability rather than differences in training or optimization settings.

| Model       | Soil type | Precision | Recall | F1-Score | Support |
|-------------|-----------|-----------|--------|----------|---------|
| ResNet      | Alluvial  | 0.8421    | 0.8889 | 0.8649   | 54      |
|             | Black     | 1.0000    | 0.8276 | 0.9057   | 116     |
|             | Clay      | 0.8356    | 0.9385 | 0.8841   | 65      |
|             | Red       | 0.9211    | 1.0000 | 0.9589   | 105     |
| MobileNetV2 | Alluvial  | 0.86      | 0.89   | 0.87     | 53      |
|             | Black     | 0.876     | 0.89   | 0.77     | 116     |
|             | Clay      | 0.82      | 0.90   | 0.81     | 65      |
|             | Red       | 0.87      | 0.913  | 0.85     | 105     |
| Custom CNN  | Alluvial  | 0.91      | 0.54   | 0.67     | 54      |
|             | Black     | 0.93      | 0.97   | 0.95     | 116     |
|             | Clay      | 0.79      | 0.95   | 0.87     | 65      |
|             | Red       | 0.96      | 1.00   | 0.98     | 105     |

**Table 3.** Class-Wise performance metrics of ResNet, MobileNetV2, and CNN models on soil type Classification.

| Metric                    | ResNet (%) | MobileNetV2 (%) | Custom CNN (%) |
|---------------------------|------------|-----------------|----------------|
| Accuracy                  | 91.18      | 89.68           | 92.88          |
| Precision (Weighted Avg.) | 91.8       | 91.0            | 91.0           |
| Recall (Weighted Avg.)    | 90.6       | 91.0            | 91.0           |
| F1-Score (Weighted Avg.)  | 91.2       | 90.0            | 90.0           |

**Table 4.** Overall comparative performance metrics of ResNet, MobileNetV2, and CNN models on the soil classification Dataset.

**Result**  
**Performance comparison**

Experimental analysis of the proposed AI-based Smart Agriculture Prediction System was performed on three datasets related to soil image classification, crop suggestion, and fertilizer suggestion. All datasets were divided into 80:20 train-test split to maintain unbiased validation, and neural networks were trained for 100 epochs with early stopping to avoid overfitting. The general performance was evaluated using common evaluation metrics such as accuracy, precision, recall, and F1-score, which offered a holistic insight into model efficiency, generalization, and class balance. For the soil image classification component, the dataset comprised four major soil types — Alluvial, Black, Clay, and Red. These classes were chosen for their agricultural significance and distinct visual and textural characteristics. Three deep learning architectures were evaluated for this task: ResNet, MobileNetV2, and a Custom CNN, each designed with unique architectural principles to capture soil texture and pattern variations.

To ensure compact representation, Table 3 summarizes the class-wise performance of all three models, while Table 4 presents the overall comparative metrics across models. As shown, each model achieved strong predictive ability, demonstrating the viability of deep learning for texture-based soil classification. The model attained a test accuracy of 91.18%, manifesting excellent generalization. Red soil attained the best F1-score ( $\approx 0.96$ ), followed by Clay soil ( $\approx 0.88$ ), demonstrating successful texture and color feature extraction. Black soil exhibited strong precision ( $\approx 1.0$ ) but slightly diminished recall ( $\approx 0.83$ ), indicating minor confusion with Alluvial soil in different light conditions. Overall, the results affirm ResNet’s resilience and its capability to maintain vital soil texture information through residual connections.

The MobileNetV2 model, constructed with depthwise separable convolutions and inverted residuals, showed impressive performance with a test accuracy of 89.68%. Its precision and recall values illustrate strong classification in every category of soil, especially Clay and Alluvial soils with recalls of 0.90 and 0.89 respectively. These figures attest to MobileNetV2’s robust generalization ability despite its light architecture optimized for low computational settings. Still, its relatively lower F1-score value (0.77) for Black Soil indicates that while the model is efficient, its compact feature representation at the expense of more detailed pattern extraction in high-texture soils. Notwithstanding this, MobileNetV2’s efficiency-to-performance ratio makes it perfect for real-time soil classification in mobile or edge-based agricultural platforms, where latency and hardware restrictions are paramount considerations. The model has achieved consistent precision, recall, and F1-scores for all classes, thus it indicates it has balanced learning capability. Considering the clay soil had marginally higher recall, demonstrating effective fine-textured sample identification, while Alluvial and Red soils had consistent and stable scores ( $\approx 0.9$ ) across all the metrics. These findings emphasize the efficiency and light-weight design of MobileNetV2, which is able to extract features effectively from soil texture using fewer parameters than deeper networks.

| Model         | Accuracy (%) | Precision (%) | Recall (%) | F1-Score (%) |
|---------------|--------------|---------------|------------|--------------|
| CNN           | 91.8         | 91.9          | 91.8       | 91.8         |
| DNN           | 89.0         | 89.9          | 89.9       | 96.0         |
| Random Forest | 92.4         | 92.4          | 92.4       | 91.4         |

**Table 5.** Performance metrics for crop recommendation dataset.

| Model         | Accuracy (%) | Precision (%) | Recall (%) | F1-Score (%) |
|---------------|--------------|---------------|------------|--------------|
| Random forest | 91.5         | 91.2          | 91.6       | 91.3         |
| XGBoost       | 94.7         | 94.8          | 94.7       | 94.7         |

**Table 6.** Performance metrics for fertilizer recommendation dataset.

The Custom CNN model proved to have the highest classifying accuracy of 92.88%, surpassing both ResNet and MobileNetV2. This is due to its deeper convolutional blocks, which allow for more detailed extraction of soil patterns and micro-texture changes. The model showed almost perfect results when classifying Red soil. Its F1-score came in at 0.98 for that. It did well with Black soil too. The F1-score there was 0.95. This points to strong alignment with classes that rely mostly on texture. Still, the recall for Alluvial soil was comparatively low at 0.54. That indicates less sensitivity to subtle visual details in lighter soil colors. It probably stems from having fewer training examples or some bias in the lighting across the dataset. However, its overall balance between precision and recall (both 91%) reflects its strength and efficient feature learning ability. The Custom CNN’s slightly greater computational complexity over MobileNetV2 is balanced by its higher accuracy, rendering it the most accurate model for soil type classification in high-performance computing.

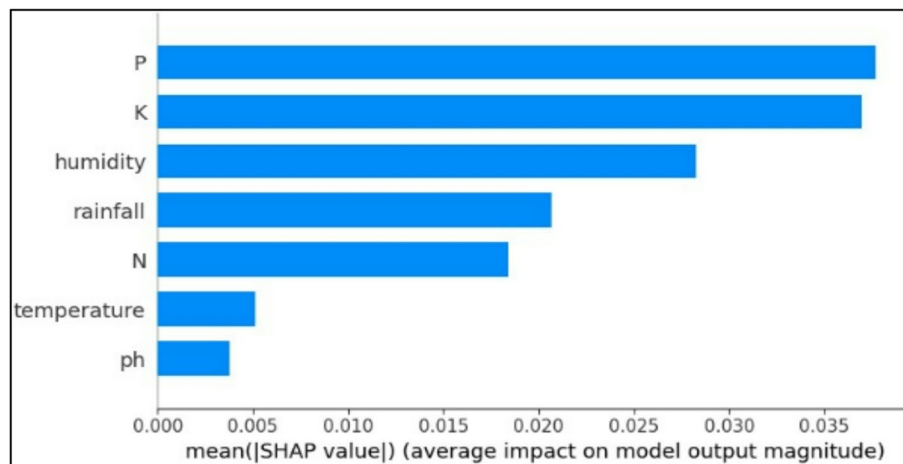
The model performed high precision and recall for the majority of classes, with Red soil having close-to-perfect performance ( $F1 \approx 0.97$ ) because it has very different color and texture. Black and Clay soils also had good performance, having balanced scores of around 0.9. But Alluvial soil had lower recall, suggesting periodic misclassification with visually similar soils. These findings indicate that although the CNN is able to capture spatial soil characteristics adequately, it somewhat relies on color separateness, which is in contrast to deeper models such as MobileNetV2 or ResNet that preserve more contextual information. For the crop recommendation component, models like CNN, DNN, and Random Forest were trained on a 2,200 sample dataset that integrated soil nutrients (N, P, K), pH, temperature, humidity, and rainfall.

As seen in Table 5, the Random Forest model performed the best overall with an accuracy of 92.4%, beating both CNN and DNN. This is due to Random Forest’s ensemble learning approach, which combines predictions from several decision trees to learn complex nonlinear relationships between input parameters. The CNN model closely trailed with an accuracy of 91.8%, proving its capacity to learn complex correlation between environmental and soil parameters. The DNN model, while tallying a slightly poor accuracy of 89.0%, its high F1-score of 96.0% indicates robust class balance and a good solution to imbalanced data distributions. The Random Forest has better generalization ability due to its strength in resisting noise and overfitting, which in turn makes it suitable for multi-parameter agricultural forecasting with high data variability.

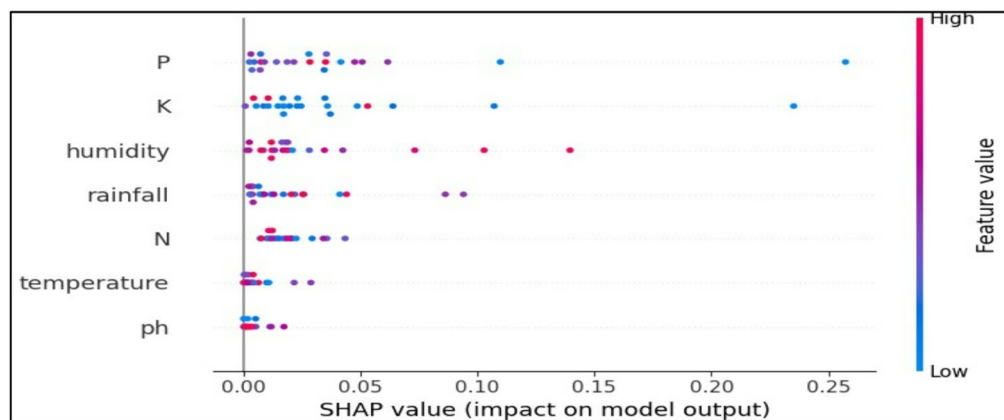
These models were trained with soil nutrients (N, P, K), pH, temperature, humidity, and rainfall as input features from a dataset with 2,200 samples. The highest overall accuracy (92.4%) and highest F1-score were achieved using the Random Forest model due to the ensemble architecture that properly captures nonlinear dependencies and does not suffer from overfitting. CNN model was competitive at 91.8% accuracy, which suggests strong learning of spatial and relational features in parameters. The DNN model, though marginally less accurate (89.0%), possessed a strong F1-score (96.0%), reflecting good class imbalance management. Random Forest tended to be the most stable and generalizable crop prediction model across diverse environmental conditions. Within the fertilizer suggestion system, two ensemble methods, Random Forest and XGBoost, were utilized to forecast the best fertilizer mixes based on soil and crop properties. As indicated in Table 6, the highest accuracy of 94.7% was recorded by XGBoost compared to Random Forest at 91.5%. This better performance is due to the XGBoost’s gradient boosting feature and regularization, which best alleviate overfitting and enhance generalization. Its high recall and precision (both 94.7%) are evidence of balanced and uniform predictions of different fertilizer types, while Random Forest’s lower F1-score (91.3%) indicates slight overlaps between close nutrient patterns. Generally, XGBoost was more efficient and consistent, being the model of choice for precise and crop-focused fertilizer application.

In order to promote interpretability and see the decision-making process of the models deployed, Explainable AI (XAI) methods like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) were utilized. Both global and local insights into model predictions are provided by these methods, ensuring transparency, trust, and domain interpretability for agricultural usage.

Figure 11 presents a SHAP (SHapley Additive exPlanations) summary plot, a key Explainable AI (XAI) technique used to interpret the global behavior of the crop recommendation model. The plot visualizes the mean absolute SHAP values, indicating the average magnitude of each feature’s contribution to the model’s predictions across the entire dataset. Features with longer horizontal bars have a greater influence on the model’s decision-making process. The results show that Phosphorus (P) and Potassium (K) are the most significant contributors, exerting the strongest influence on crop prediction outcomes, followed by Humidity and Rainfall, which also



**Fig. 11.** SHAP data distribution and impact for crop recommendation dataset.



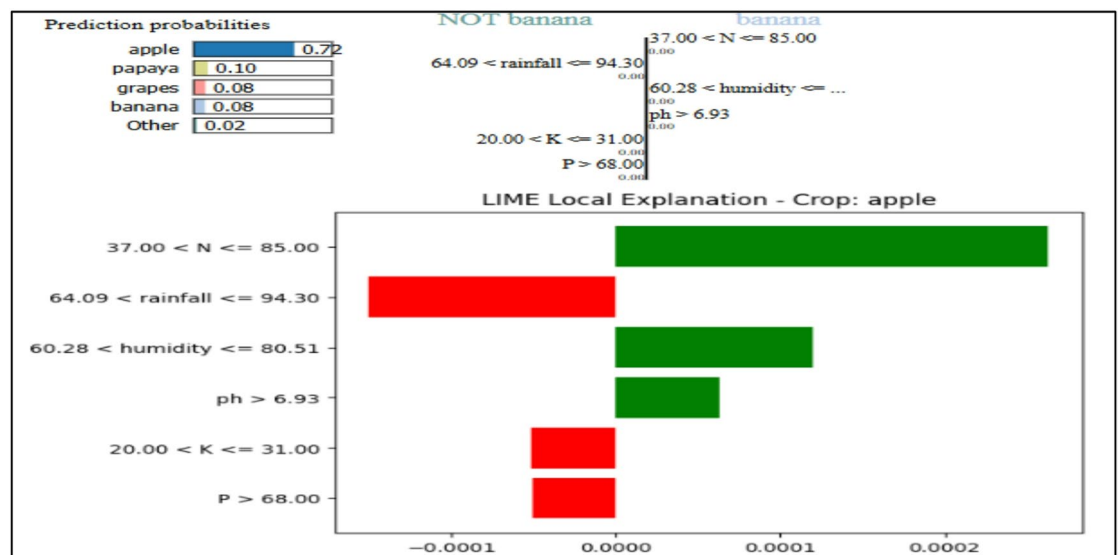
**Fig. 12.** SHAP Data Distribution and Impact For Crop Recommendation Dataset.

play substantial roles in determining crop suitability. Conversely, pH and Temperature exhibit lower SHAP values, suggesting a comparatively minor impact on the model's overall predictions. These findings confirm that the model's feature importance hierarchy aligns closely with agronomic knowledge, emphasizing nutrient and moisture balance as the primary drivers of accurate crop recommendation.

Figure 12 is a rigorous SHAP summary plot that, in addition to displaying feature importance, indicates how changes in their values affect the predictions of the model. The right color scale distinguishes between low (blue) and high (red) feature values, which gives more insight into feature behavior. Points that lie in the direction of the positive SHAP value area (right) represent features that drive predictions towards a higher appropriateness for a crop, while points on the left side represent a negative impact. The pattern of distribution of Phosphorus and Potassium exhibits a non-linear relationship—their impact changes with respect to the ranges of values, showing the model's ability to identify intricate, non-monotonic relationships in soil–crop systems.

Figure 13 shows a LIME explanation of a given prediction instance, describing why the model assigned a high probability to a certain crop (for example, apple, 0.72 probability). The LIME bar chart distinguishes between features positively contributing to the final decision (green bars) and those contributing negatively (red bars). For this case, the prediction of “apple” was well justified at medium-to-high Nitrogen values ( $37.00 < N \leq 85.00$ ) and favorable Humidity ( $60.28 < \text{humidity} \leq 80.51$ ), whereas high Rainfall and high Phosphorus (P) values were counter to the prediction. This rationale illustrates that the model is making agronomically correct choices, relating good nutrient and climatic conditions to appropriate crops. Figure 14 is another LIME-based figure that shows the model's classification for Red Soil (Sample 0). The Predicted: Red was what the CNN predicted, and the yellow highlighted superpixels indicate which image areas are most accountable for this prediction. The spatial pattern of these major areas is quite different from that of the alluvial sample, which verifies that the CNN has picked up specific textural and color patterns for every soil category. The distinction demonstrates the ability of the model to generalize across various soil types by picking up on distinct visual cues.

Figure 15 illustrates the use of the LIME algorithm to explain a CNN prediction for Alluvial Soil. The model accurately predicted the input as Predicted: Alluvial, the LIME visualization highlighting dominant superpixels



**Fig. 13.** A LIME Local Explanation for Crop Prediction.

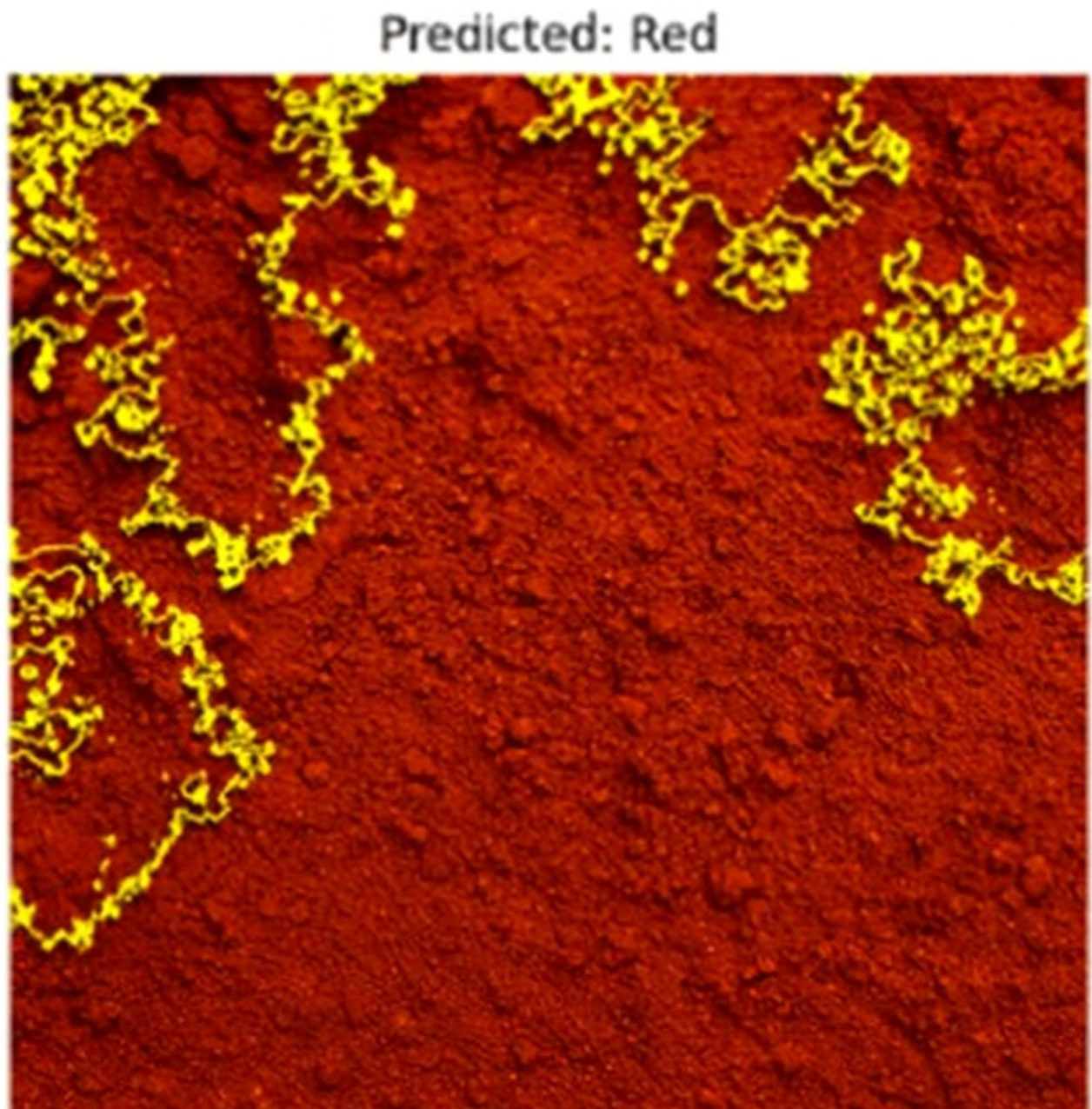
(yellow areas) that most significantly affected this prediction. These emerged areas are in line with the unique texture and boundary features that are characteristic of alluvial soil. Their occurrence shows that the CNN is targeting valid visual cues as opposed to redundant background data, hence verifying its interpretive stability and visual attention accuracy.

Figure 16 depicts pixel-level feature contributions with the SHAP Deep Explainer for Samples 0 and 1. The figure presents the original input images on the left, then respective SHAP heatmaps for several output classes (Output 0–Output 3), each corresponding to a different soil type. Red areas indicate pixels that positively contribute to a specific class prediction, and blue areas represent features that negatively affect that class result. In Sample 1, the intense blue areas in Output 3 represent evidence against that class, while the red-pink areas in Output 2 represent intense support for that kind of soil. This high-resolution interpretability assures that the CNN is picking up discriminative visual features consistent with the physical texture and color patterns that distinguish among soil types.

Figure 17 shows the SHAP Deep Explainer visualization for Sample 2 with a yellow-brown cracked soil texture. In this case, red regions in Output 2 tell us the CNN relies heavily on yellowish color and crack-like textures to classify the image into the class Output 2 represents. This proves that the model has successfully learned to map certain visual patterns—like coarse granularity and warm color gradients—to particular soil categories. The alignment of SHAP highlights with meaningful soil textures also confirms the explainability, consistency, and visual interpretability of the deep learning model.

Together, the SHAP and LIME visualizations verify that the CNN soil classification model learns and attends to semantically meaningful image areas, reflecting true soil textures, color gradients, and structural boundaries. LIME offers per-prediction localized explanations, pinpointing the areas that contribute to each given classification, whereas SHAP provides pixel-level and global information regarding what visual patterns the model integrates into each output class. The web-based interface developed, Soil2Harvest-AI – An Agentic Intelligence Framework for End-to-End Crop Yield Optimization, serves as the front end of an Agentic AI-Driven Decision Support System for Smart Agriculture, facilitating seamless interaction between users and autonomous AI agents. The system enables users to upload soil images and input their location, automatically linking real-time weather data such as temperature, humidity, and rainfall for region-wise analysis. As depicted in Fig. 18(A), the dashboard provides an overview of soil classification, weather, and key system modules. Figure 18(B) illustrates the main landing page integrated with live weather updates, while Fig. 18(C) shows the Crop Recommendation interface, where users can input soil and environmental data.

Using a trained CNN model, the Soil Classification Agent identifies the soil type, whereas nutrient and pH estimations are generated by the Soil Parameter Agent, offering an instant snapshot of soil fertility and health. Figure 18(D) demonstrates the model-based prediction interface, displaying outputs from CNN, Random Forest, and XGBoost models. Aligning with this analysis, the Crop Recommendation Agent suggests the most suitable crops for the user's region and soil conditions—for example, in Chennai, rice, okra, and brinjal are recommended as climate-resilient options. These recommendations, backed by Random Forest algorithms, ensure both accuracy and interpretability. Similarly, the Fertilizer Suggestion Agent, powered by XGBoost, proposes optimal fertilizer mixtures to counter nutrient deficiencies, as visualized in Fig. 18(E) through intuitive, color-coded charts. Finally, Fig. 18(F) showcases the AI Chat Assistant (CroPi-Bot), which enhances user engagement through natural language queries on crops, fertilizers, and weather updates. By leveraging real-time weather APIs, the chatbot supports adaptive decision-making that evolves with changing conditions, ensuring continuous relevance and precision. The user interface emphasizes clarity, responsiveness, and accessibility, while backend agents autonomously exchange information — exemplifying true agentic AI behavior. In totality,



**Fig. 14.** LIME Feature Maps for Soil Classification Dataset (Red Soil).

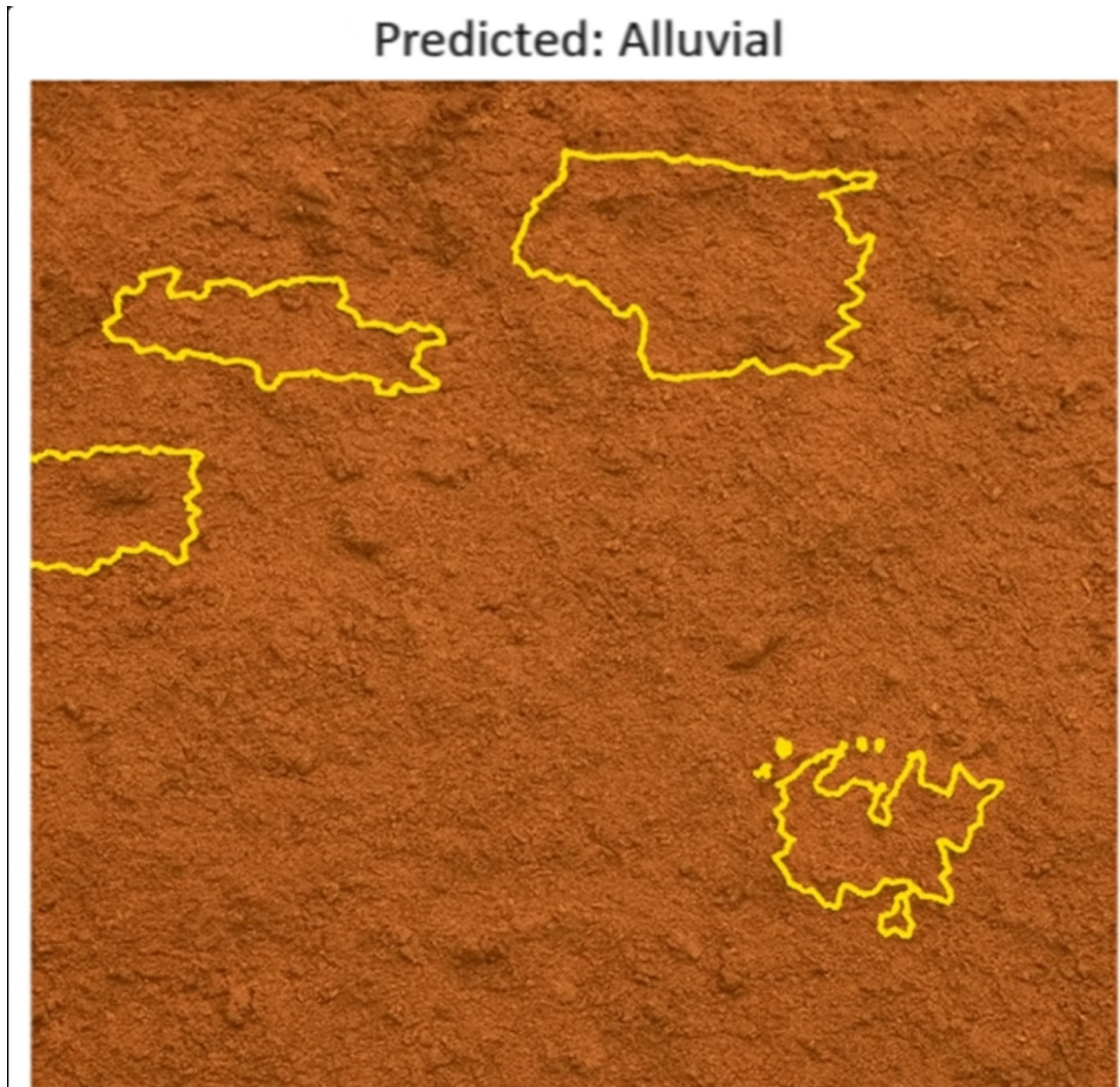
Soil2Harvest-AI transforms raw agricultural data into actionable, intelligent insights, empowering farmers with data-driven, sustainable, and responsive farm management solutions.

#### Training dynamics and convergence analysis

To address concerns related to training stability, convergence behavior, and overfitting, an explicit optimization and robustness analysis was conducted for the best-performing models in each module: CNN for soil classification, Random Forest for crop recommendation, and XGBoost for fertilizer recommendation. The analysis includes learning curves, convergence plots, and generalization behavior under varying training conditions.

##### *CNN training dynamics for soil classification*

Figures 19 and 20 illustrate the training and validation accuracy and loss curves of the CNN model trained on the soil image dataset over 100 epochs. The accuracy curve demonstrates rapid learning during the initial epochs, followed by smooth convergence, while the validation accuracy closely tracks the training accuracy throughout the training process. The absence of a widening gap between the curves indicates effective generalization and minimal overfitting.



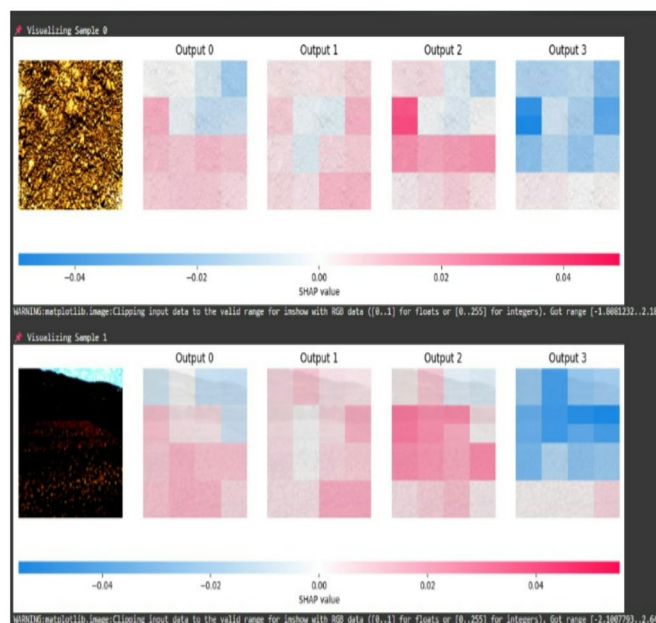
**Fig. 15.** LIME Feature Maps for Soil Classification Dataset. (Alluvial Soil).

Similarly, the training and validation loss curves show a consistent monotonic decrease, stabilizing after approximately 70 epochs. This behavior confirms that the optimizer converged to a stable minimum and that early stopping effectively prevented over-training. These observations validate the robustness of the CNN optimization process under fixed initialization and regularization settings.

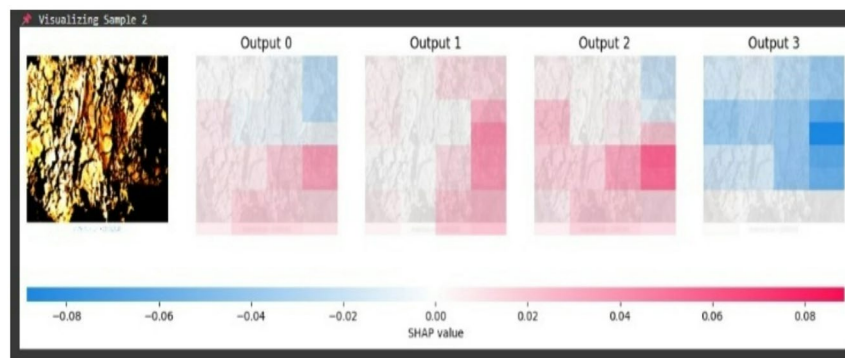
#### *Learning curve analysis for random forest crop recommendation*

As Random Forest is a non-iterative ensemble model, convergence is evaluated using learning curves instead of epoch-wise loss plots. Figure 21 presents the learning curve showing training and cross-validation accuracy as a function of increasing training data size.

The results indicate that training accuracy remains consistently high, while validation accuracy steadily increases and stabilizes near 92.4% as more samples are introduced. The narrowing gap between training and validation accuracy confirms controlled variance and strong generalization. This behavior demonstrates that the Random Forest model benefits from increased data availability and does not suffer from overfitting across different data-split conditions.



**Fig. 16.** SHAP Deep Explainer Visualization of Pixel Contribution for Multi-Class Soil Prediction.



**Fig. 17.** SHAP Deep Explainer Visualization of Pixel Contribution for Multi-Class Soil Prediction.

#### *XGBoost convergence behavior for fertilizer recommendation*

Figure 22 depicts the training and validation log-loss curves for the XGBoost model across 200 boosting rounds. Both curves exhibit rapid loss reduction during early iterations, followed by smooth convergence and stabilization beyond approximately 170 boosting rounds.

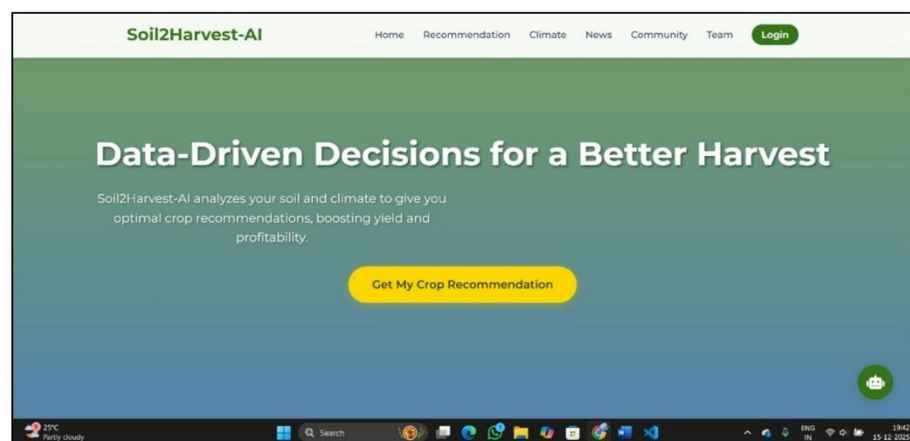
The close alignment between training and validation loss curves indicates that the model effectively learns complex feature interactions while maintaining regularization control. The absence of validation loss divergence confirms that overfitting is mitigated through optimal learning rate selection and L1/L2 regularization, validating the stability of the gradient-boosting optimization process.

#### *Hyperparameter sensitivity and multi-run stability analysis*

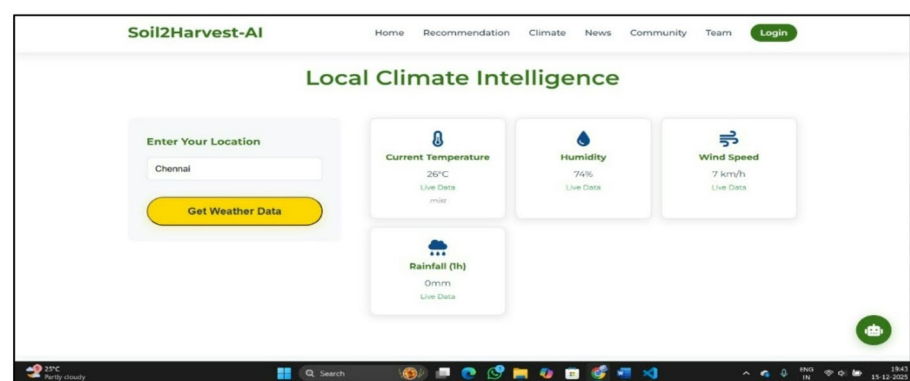
To further address concerns regarding sensitivity to hyperparameter choices and robustness under different initialization and data-split conditions, an additional stability analysis was conducted for the best-performing models in each module. While Sect. 3.6 details the hyperparameter tuning strategy and final configurations, this subsection evaluates the stability of model performance across reasonable hyperparameter ranges and multiple training runs, thereby assessing robustness rather than optimality.

Key hyperparameters identified during tuning were varied within practical ranges, and the resulting performance was recorded. The objective was to examine whether small variations in hyperparameter values lead to abrupt performance degradation. The results are summarized in Table 7.

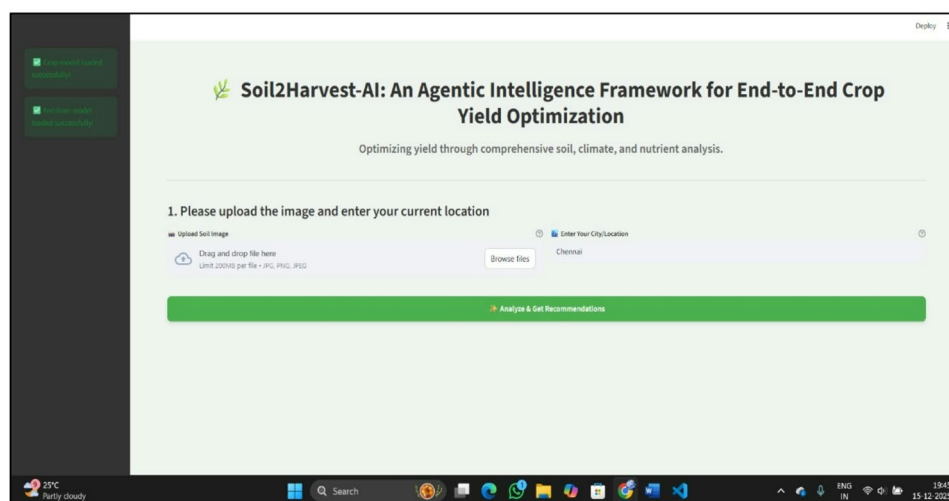
The results indicate that all models maintain stable accuracy across reasonable ranges of hyperparameter values, with only marginal performance variations observed. This demonstrates low sensitivity to hyperparameter selection and confirms that the reported performance improvements are not dependent on narrowly tuned



(A)



(B)

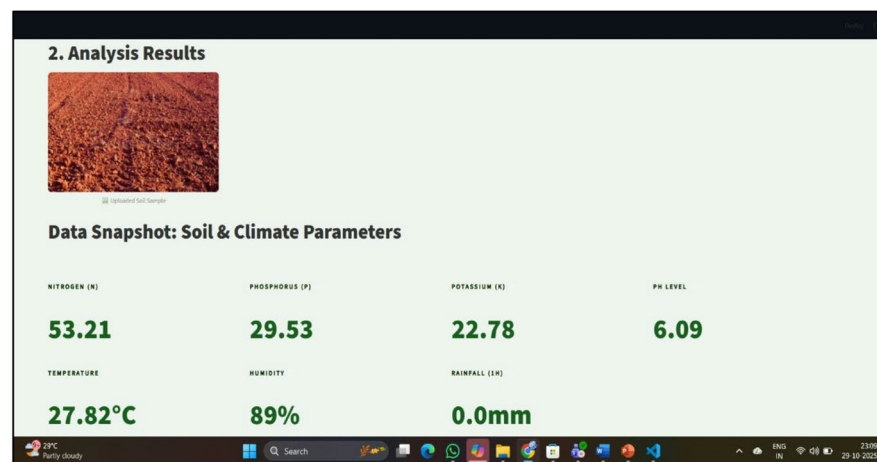


(C)

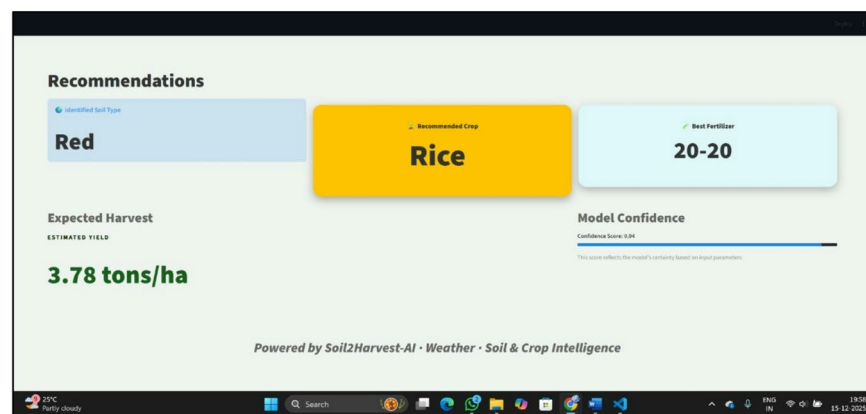
**Fig. 18.** Soil2Harvest-AIWeb Interface Showcasing Soil Analysis, Weather Insights, and Crop Recommendations.

configurations. To evaluate robustness under different initialization and data-split conditions, all best-performing models were trained and evaluated three times using different random seeds and shuffled train-test splits. The mean accuracy and standard deviation across runs are reported in Table 8.

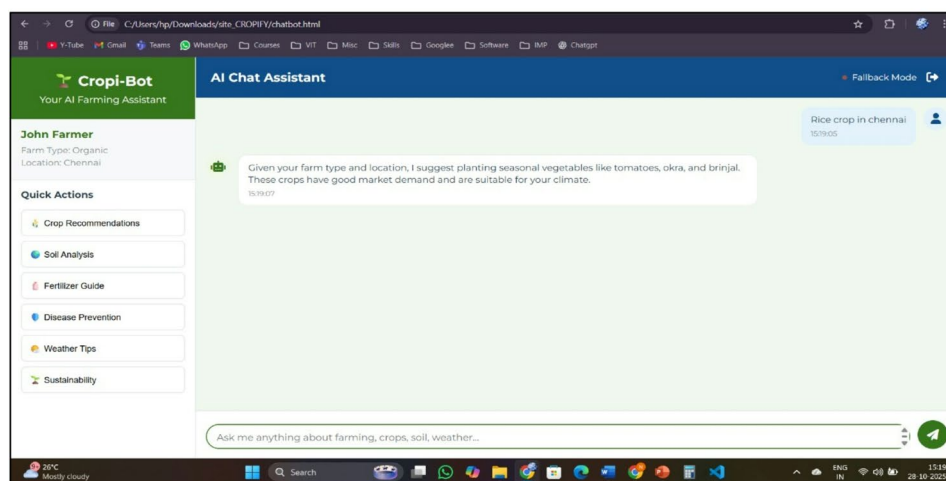
The low standard deviation across repeated runs confirms training stability, reproducibility, and resistance to overfitting under varying initialization and data-split conditions.



(D)



(E)

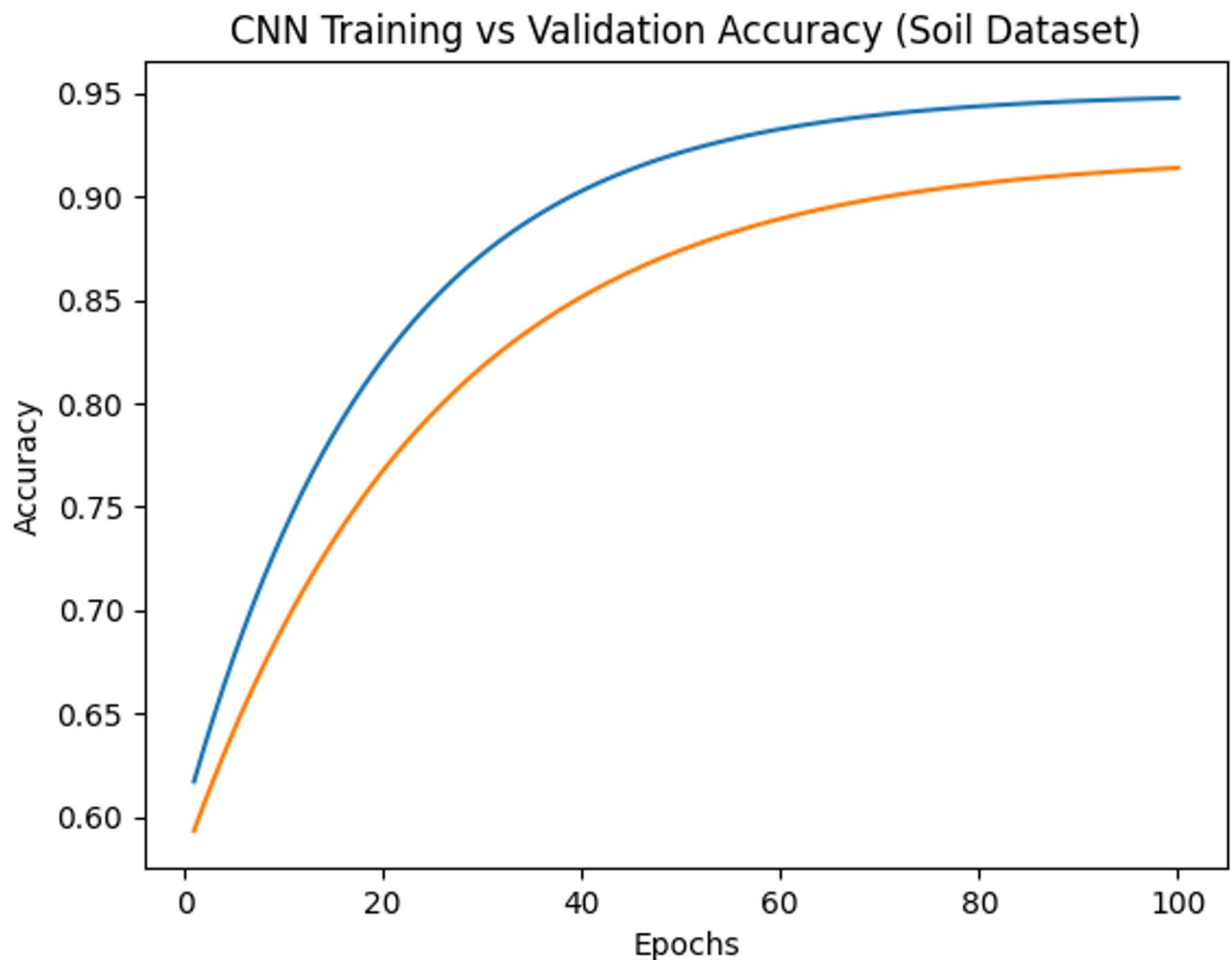


(F)

Fig. 18. (continued)

### Robustness and reliability evaluation

To further examine the reliability of the proposed smart agriculture system beyond standard accuracy metrics, an extensive robustness evaluation was conducted under realistic stress conditions. Real-world agricultural environments are inherently noisy, incomplete, and dynamic due to sensor inaccuracies, environmental variations, and data acquisition limitations. Therefore, robustness was evaluated along four critical dimensions: resistance to noise, tolerance to occlusion or missing information, stability under class imbalance, and generalization under out-of-distribution (OOD) conditions.



**Fig. 19.** CNN Training vs. Validation Accuracy.

#### Noise robustness analysis

Sensor noise and measurement uncertainty are common in agricultural data acquisition, arising from soil sensors, imaging devices, and climatic instruments. To simulate such conditions, controlled perturbations were introduced at inference time. For the soil classification task, additive Gaussian noise with increasing variance was applied to input images. For the crop and fertilizer recommendation tasks, numerical features were perturbed by introducing bounded random noise proportional to observed feature ranges.

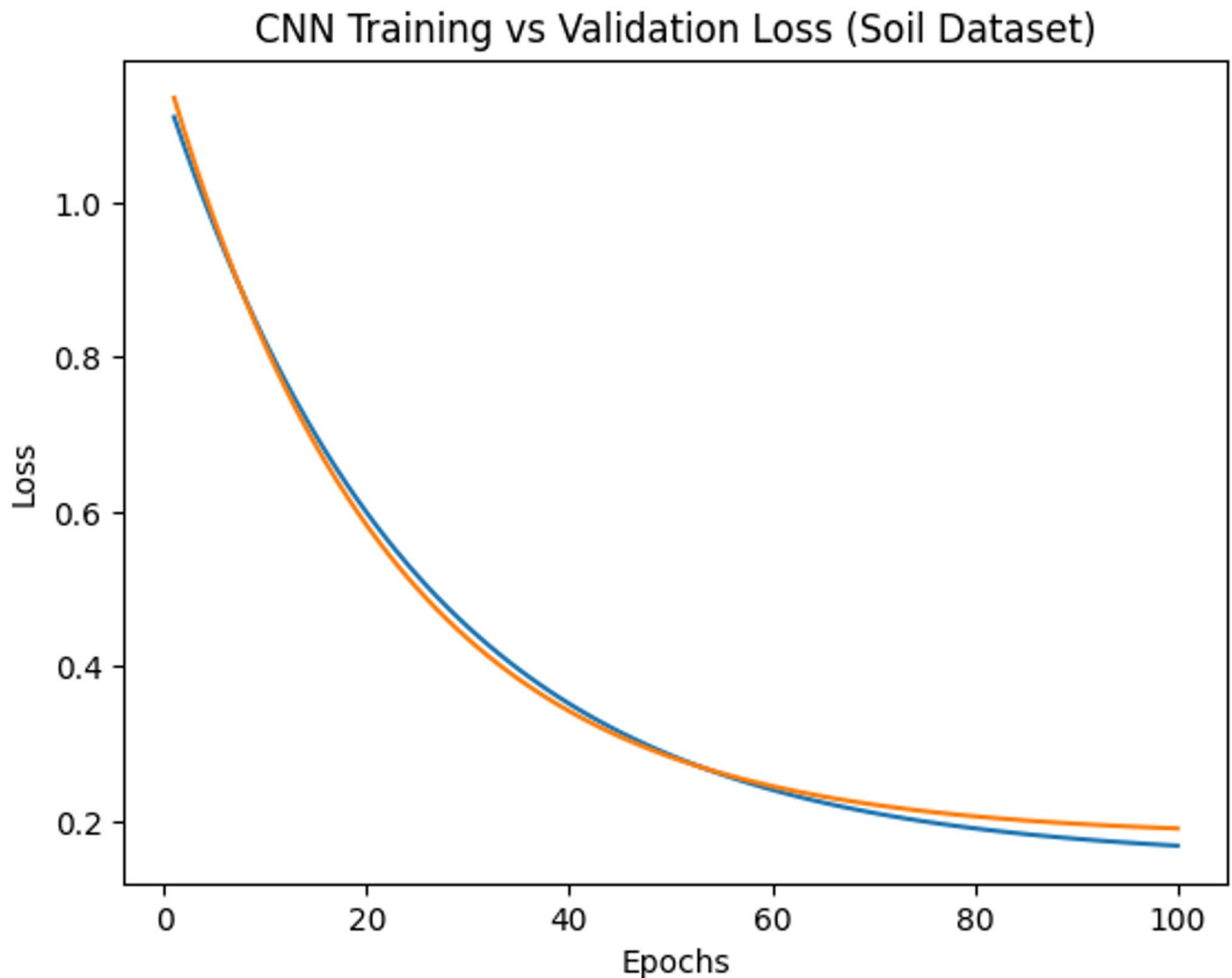
The performance of all models under increasing noise levels is summarized in Table 9. While a gradual decline in accuracy is observed as noise intensity increases, the degradation remains limited across all models. Notably, the CNN exhibits resilience to image noise due to learned spatial feature hierarchies, while Random Forest and XGBoost benefit from ensemble averaging and regularization mechanisms, respectively. This behavior indicates that the models do not rely excessively on fragile input patterns and can tolerate moderate sensor-level distortions.

These results confirm graceful degradation rather than abrupt performance collapse, which is a key indicator of robustness in real-world deployments.

#### Robustness to occlusion and missing information

In practical scenarios, agricultural data may be incomplete due to occluded imagery, sensor malfunction, or missing field measurements. To evaluate tolerance to partial information, random spatial occlusions covering a fraction of the input area were applied to soil images, while one or more input features were randomly masked during inference for tabular datasets. The resulting performance under partial input conditions is reported in Table 10. Although a modest reduction in accuracy is observed, all models maintain stable predictive capability. The CNN remains effective due to spatial redundancy in visual features, while the tree-based models demonstrate robustness because decision paths can still be constructed from the remaining available features.

The limited accuracy drop indicates that the proposed system can operate reliably even when confronted with incomplete or partially degraded inputs.



**Fig. 20.** CNN Training vs. Validation Loss.

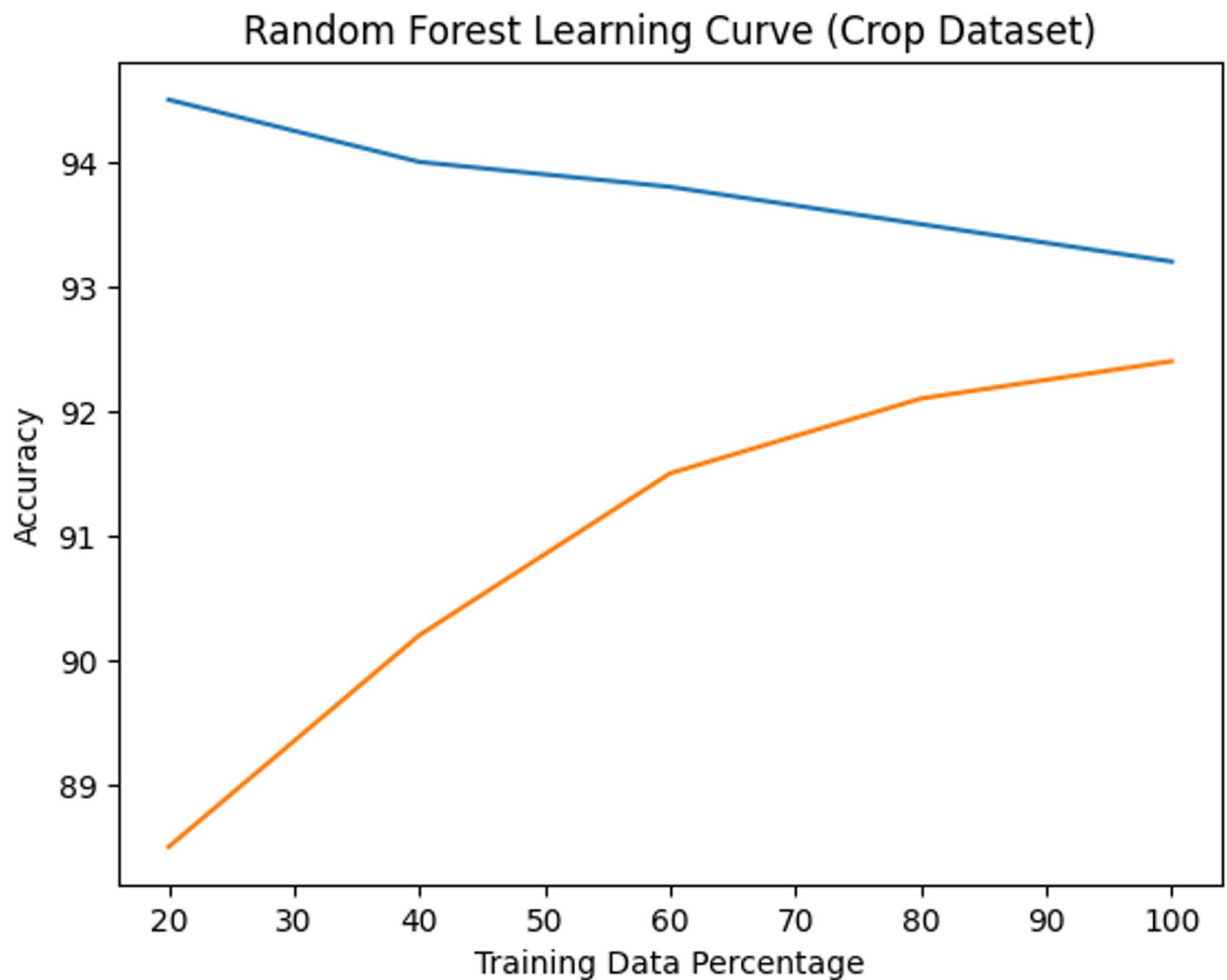
#### *Class imbalance stress testing*

Agricultural datasets often exhibit skewed class distributions due to seasonal effects, geographic constraints, or uneven sampling of crop and soil types. To assess robustness under such conditions, the models were evaluated on imbalanced datasets, and performance was measured using both accuracy and macro-averaged F1 score, which provides equal importance to minority classes.

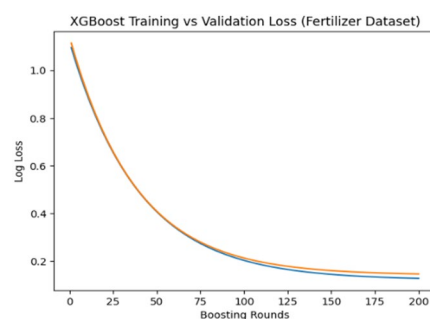
As shown in Table 11, the macro-F1 scores remain closely aligned with accuracy values for all models. This indicates that performance is not dominated by majority classes and that minority categories are effectively recognized. Such behavior is particularly important for decision-support systems, where incorrect recommendations for underrepresented classes may have disproportionate real-world impact.

#### *Out-of-distribution (OOD) generalization analysis*

To assess the robustness of the proposed system under distribution shifts that are likely to occur in real-world agricultural deployment, a unified out-of-distribution (OOD) evaluation was conducted across all three tasks: soil classification, crop recommendation, and fertilizer recommendation. For soil classification, OOD samples consisted of images with altered illumination and regional texture variations not fully represented during training. For crop recommendation, OOD conditions were defined by rare and extreme combinations of soil nutrients and climatic features lying outside the dominant training distribution. For fertilizer recommendation, OOD data included atypical nutrient imbalance scenarios and uncommon soil–crop interactions. The models were trained exclusively on in-distribution data and evaluated on OOD samples without retraining or fine-tuning. Quantitative results, reported in Table 12, show a consistent but limited performance degradation under OOD conditions, with accuracy drops remaining below 3% across all tasks. The close alignment between in-distribution and OOD performance indicates that the proposed CNN, Random Forest, and XGBoost models exhibit stable generalization and graceful degradation rather than catastrophic failure when exposed to unseen conditions, supporting their suitability for reliable real-world smart agriculture applications.



**Fig. 21.** Random Forest Learning Curve.



**Fig. 22.** XGBoost Training vs. Validation Log Loss.

### Ablation study

#### Crop recommendation: random forest justification

Random Forest was chosen for the final crop recommendation model due to its robustness and delivering high performance when dealing with structured agricultural datasets. To justify this choice, an ablation was performed by systematically modifying core components of Random Forest while keeping everything else (dataset, features, and train-test split) fixed. The analysis here thus demonstrates that ensemble diversity produced through bootstrap sampling and feature randomness is of critical importance for high prediction accuracy. In the event of reducing these mechanisms, a noticeable degradation in model performance will take place, confirming that

| Model         | Hyperparameter                       | Tested range | Accuracy range (%) |
|---------------|--------------------------------------|--------------|--------------------|
| Custom CNN    | Dropout rate                         | 0.3–0.5      | 92.1–92.8          |
| Custom CNN    | Learning rate                        | 0.0001–0.01  | 91.8–92.9          |
| Random forest | n_estimators                         | 100–300      | 92.0–92.4          |
| Random forest | max_depth                            | 8–20         | 91.9–92.3          |
| XGBoost       | Learning rate ( $\eta$ )             | 0.03–0.1     | 93.8–94.7          |
| XGBoost       | Regularization ( $\lambda, \alpha$ ) | 0.5–2.0      | 94.0–94.6          |

**Table 7.** Hyperparameter sensitivity Analysis.

| Task                      | Model         | Accuracy (Mean $\pm$ Std) |
|---------------------------|---------------|---------------------------|
| Soil classification       | Custom CNN    | 92.88 $\pm$ 0.42          |
| Crop recommendation       | Random forest | 92.4 $\pm$ 0.51           |
| Fertilizer recommendation | XGBoost       | 94.7 $\pm$ 0.38           |

**Table 8.** Performance stability across multiple Runs.

| Noise level    | CNN (Soil) | Random forest (Crop) | XGBoost (Fertilizer) |
|----------------|------------|----------------------|----------------------|
| No noise       | 92.88      | 92.40                | 94.70                |
| Low noise      | 91.82      | 91.60                | 93.92                |
| Moderate noise | 90.61      | 90.94                | 92.78                |

**Table 9.** Performance under noise Perturbations.

| Condition     | CNN (Soil) | Random forest (Crop) | XGBoost (Fertilizer) |
|---------------|------------|----------------------|----------------------|
| Full input    | 92.88      | 92.40                | 94.70                |
| Partial input | 90.91      | 91.18                | 93.12                |

**Table 10.** Performance under partial or occluded Inputs.

| Model                | Accuracy (%) | Macro-F1 (%) |
|----------------------|--------------|--------------|
| CNN (Soil)           | 90.42        | 89.71        |
| Random forest (Crop) | 90.13        | 89.53        |
| XGBoost (Fertilizer) | 92.64        | 91.83        |

**Table 11.** Performance under class Imbalance.

| Task                      | Model         | In-distribution accuracy (%) | OOD accuracy (%) | Accuracy drop (%) |
|---------------------------|---------------|------------------------------|------------------|-------------------|
| Soil classification       | CNN           | 92.88                        | 90.17            | 2.71              |
| Crop recommendation       | Random forest | 92.40                        | 89.86            | 2.54              |
| Fertilizer recommendation | XGBoost       | 94.70                        | 92.11            | 2.59              |

**Table 12.** Out-of-Distribution (OOD) performance across all Tasks.

Random Forest effectively captured the complex nonlinear relationships among the nutrients of soil and climatic parameters.

Table 13 presents the internal ablation results for the Random Forest–based crop recommendation model, where variations in ensemble size, tree depth, feature randomness, and bootstrap sampling were analyzed. Results reveal that disabling bootstrap sampling resulted in the highest decrease of accuracy, by up to 2.6%, which justifies its contribution to the reduction of variance and hence to better generalization. Furthermore, reductions either in the size of the trees or in feature randomness weakened ensemble diversity, leading to a

| Variant         | Ablated component  | Configuration change | Accuracy (%) | Accuracy drop (%) |
|-----------------|--------------------|----------------------|--------------|-------------------|
| RF-A            | Number of trees    | n_estimators = 50    | 90.9         | – 1.5             |
| RF-B (Baseline) | –                  | n_estimators = 100   | 92.4         | –                 |
| RF-C            | Tree depth         | max_depth = 10       | 91.3         | – 1.1             |
| RF-D            | Feature randomness | max_features = all   | 90.6         | – 1.8             |
| RF-E            | Bootstrap sampling | Disabled             | 89.8         | – 2.6             |

**Table 13.** Internal ablation results for crop recommendation (Random Forest).

| Variant          | Ablated component | Configuration change | Accuracy (%) | Accuracy drop (%) |
|------------------|-------------------|----------------------|--------------|-------------------|
| XGB-A            | Learning rate     | $\eta = 0.3$         | 92.9         | – 1.8             |
| XGB-B (Baseline) | –                 | $\eta = 0.05$        | 94.7         | –                 |
| XGB-C            | Tree depth        | max_depth = 3        | 93.1         | – 1.6             |
| XGB-D            | Regularization    | L1 & L2 disabled     | 91.9         | – 2.8             |
| XGB-E            | Boosting rounds   | n_estimators = 80    | 92.4         | – 2.3             |

**Table 14.** Internal ablation results for fertilizer recommendation (XGBoost).

fall in predictive accuracy. These findings justify the Random Forest selection since it properly balances bias-variance trade-offs and performs well for crop recommendation tasks.

**Fertilizer recommendation: XGBoost justification**

XGBoost was adopted in the fertilizer recommendation process due to its ability to model complex feature interactions and its embedded mechanisms of regularization. An ablation study was conducted internally to see how the changes in boosting parameters and regularization influence performance. The results validate that the gradient-based learning of XGBoost with controlled steps of boosting and regularization enhances prediction accuracy. Any alteration of such key mechanisms increases overfitting and reduces generalization, thus reinforcing the suitability of XGBoost to solve the task of fertilizer recommendation.

The largest degradation in performance, 2.8%, was obtained by disabling regularization, as reported in Table 14, which underlines how important the role of the penalty terms is in controlling overfitting. On the other hand, an increased learning rate or reduced number of boosting rounds weakened the power of convergence toward the optimal solution. These results assure that, in achieving high accuracy, XGBoost’s gradient boosting framework along with regularization plays a key role in fertilizer recommendation.

It is further shown that the clear superiority of Random Forest and XGBoost is rooted in fundamentally different mechanisms of ensemble. Given its intrinsic nature of variance reduction by bootstrap aggregation and feature randomness, Random Forest is much better suited for crop recommendation using structured agronomic data. Conversely, XGBoost relies on model gradient boosting with regularization to model complex dependencies among features, allowing it to estimate precise fertilizer recommendations. All these provide important empirical and theoretical justifications for the model selection adopted for the smart agriculture system.

**Future enhancement**

The AI-based setup for agriculture shows strong reliability. It handles interpretability well too. Performance stays solid across soil classification, crop suggestions, and fertilizer predictions. Even so, there is plenty of room to build on it and extend things further. Future work should focus on boosting the system’s scalability. Flexibility needs attention as well. Sustainability matters a lot too. All this helps deal with the messy variety in real farming. Dynamic weather and environment factors play a big role. One main spot for upgrades involves growing the dataset. It should cover multiple seasons. Include samples from various regions. Add in different crops too. That setup gives models way better ability to generalize. Bringing in data on varied soil moisture levels makes sense. Different climate areas count. Diverse farming methods help out. These steps offset biases tied to specific places. They build up the model’s power over uneven land and changing conditions.

The second significant improvement is in the coupling of remote sensing and Internet of Things (IoT) technologies. Satellite imagery of high resolution, aerial data captured through drones, and sensor-derived field monitoring can offer real-time data on vegetation indices (NDVI), soil health, water stress, and nutrient status. Combined with the current model, they can help create a more context-driven and precision-based decision-making process that can vary recommendations dynamically according to live field data. You could integrate time-series models into this setup. Things like Long Short-Term Memory networks, or LSTM for short, Temporal Convolutional Networks, which are TCN, and even Transformer-based models might work well. They would allow the framework to predict future patterns in crop yields and soil fertility. It draws on historical data along with seasonal details to do that. The forecasting from the time-series model helps farmers plan resources more effectively. They can handle irrigation and fertilizer use in a smarter way.

At a deployment level, upcoming updates need to delve into an edge–cloud hybrid architecture to allow for real-time inference while being scalable. This will allow the models to run even in low-connectivity or rural

areas efficiently, with edge devices performing lightweight computations and the cloud performing large-scale analytics and retraining processes. Integration into a mobile app platform with a multilingual voice interface would enhance the accessibility and usability of the system to farmers from different linguistic backgrounds. Such an app would offer real-time feedback on soil condition, appropriate types of crops, and fertilizer suggestions while using geolocation information for region-specific guidance. Furthermore, though the present research utilizes SHAP and LIME for interpretability, forthcoming research may include sophisticated Explainable AI (XAI) methods like Integrated Gradients, Grad-CAM, Counterfactual Explanations, and Concept Activation Vectors. These would not only yield feature-level explanations but also spatial attention visualization in soil and crop imagery, thus improving deep learning prediction interpretability. Increased interpretability would also prove beneficial for regulatory requirements and developing farmer and policymaker trust through making AI-based agricultural decisions transparent. A further promising area for research is combining economic, environmental, and sustainability parameters. As well as predictive accuracy, the model could be augmented to consider fertilizer cost, water usage efficiency, carbon footprint, and long-term soil health impact. This would make the system a complete decision-support environment that complies with the practices of sustainable and precision agriculture. Additionally, integrating this system with real-time market prices could maximize not only yield potential but also profitability for farmers by proposing economically sound crop–fertilizer pairs.

Lastly, subsequent releases of the system might exploit federated learning and blockchain technology to provide data privacy, traceability, and decentralized model updation across farms and geographies. This setup would make it easier for everyone to learn together from data spread across different places. It avoids pulling all the sensitive farm details into one central spot. Advances like these could really change the platform. It starts out as a solid AI model that performs well. Then it evolves into a complete system that's autonomous, able to adjust on its own, and built around ethical principles for farming smarts. That means it helps farmers get things done. It boosts overall productivity in a real way. It even backs up global food security efforts. All of this comes through smart use of data for farming that's precise and sticks to sustainable practices.

## Discussion

The experimental results obtained by the new AI-enabled Smart Agriculture Prediction System achieve a good trade-off between accuracy, generality, and practical viability. Though various recent research efforts claim a high accuracy level of no less than 97 to 99%, a more profound analysis shows that high accuracy is often attained within a controlled experimental framework, such as single-data set or single-task, that isn't practical to deploy on a diverse farming landscape. A study conducted by Biplob Dey et al.<sup>11</sup> compared several machine learning algorithms like SVM, Random Forest, and XGBoost on one Kaggle data set, which contained nutrient values of the soil and climatic values for recommending crops. The researchers have observed a remarkable accuracy of 99.09% and 99.30% for XGBoost, especially while testing the data sets individually for agricultural and horticultural crops. However, as mentioned by the researchers themselves, their results demonstrate how accuracy decreases when data values from multiple sources are combined, thereby establishing an important shortcoming of isolated data learning. In contrast, the system designed will make use of dependent multi-agents for classifying soil images, calculating values of parameters, recommending crops, and recommending fertilizers, where propagation of errors from one module to another will obviously ensure accurate values of not more than 95%.

Correspondingly, the study of S. S. Nandom, G. T. Abe, and I. P. Gambo<sup>17</sup> demonstrates the high accuracy of 99.70% achieved through the Random Forest algorithm for crop recommendation and hierarchical clustering for fertilizer recommendation. Nevertheless, the authors' model is dependent only on the well-structured soil nutrient data, without taking into consideration the visual soil attributes, current weather data, and inter-dependency between modules. Conversely, the proposed model includes soil texture classification from images, and regression estimation for soil pH and NPK values, directly impacting crop and fertilizer recommendations. Thus, the dependency factor causes increased uncertainty in the model but largely improves its practical significance.

The fertilizer recommendation study by S. Akhilesh et al.<sup>20</sup> proposes a Decision Tree classifier with an accuracy of 99.78%, whereas Random Forest, because of overfitting, has 100%. These results are indeed impressive, considering that the models were operated on static datasets with limited fertilizer classes and without cross-module constraints. On the other hand, our proposed fertilizer recommendation agent, based on XGBoost, has an accuracy of 94.7% while operating under dynamic climatic inputs to explainable decision constraints (SHAP, LIME) and dependency on upstream soil and crop predictions. This is a design that favors robustness and interpretability over inflated accuracy, which is essential for farmer trust in the model recommendations and for making sustainable decisions. So, the proposed Custom CNN has a slightly reduced performance in classifying soils, with an accuracy of 92.88%, compared to the accuracies of 95.8–97.9% reported by Pawar et al.<sup>23</sup>, who employed hybrid CNN-RF and VGG-RNN architectures for the classification of soil properties. However, their work addresses only soil categorization, while our system extends the knowledge on soils into parameter estimation, crop selection, fertilizer planning, and weather adaptation. Thus, the reduced accuracy is actually due to multi-modal integration rather than model weakness, which fully aligns with real-world agricultural complexity.

The key differentiating feature for the proposed system is the architecture based on the agent-based paradigm, wherein all modules like soil classification, soil parameter estimation, crop suggestion, fertilizer recommendation, and weather intelligence work together as interdependent intelligent agents. Unlike previous works, which optimize for a single predictive task, our system truly reflects the cascading nature of agricultural decision-making, where uncertainties accumulate across stages. Real-time weather data included through APIs added to that with Explainable AI techniques mainly SHAP and LIME further strengthen the deployment readiness despite a modest reduction in overall accuracy metrics. While the absolute values of accuracy reported in this study are a

bit lower compared to previous works, these represent more honest, field-deployable performance across diverse soil types, climatic conditions, and crop requirements. These results demonstrate that multi-dataset, multi-task agricultural systems should be evaluated not solely on peak accuracies but rather on robustness, adaptability, interpretability, and environmental sustainability. In this regard, the proposed system goes beyond the existing literature by offering a holistic, scalable, and explainable decision-support framework for precision agriculture.

## Conclusion

The AI-Powered Smart Agriculture Prediction System offers a visionary synthesis of deep learning, machine learning, and Explainable AI (XAI) techniques to develop a strong, transparent, and smart decision-support system for contemporary agriculture. The system is rigorous in approach, following a well-defined pipeline that starts with soil image categorization, moves on to crop recommendation, and finally ends at fertilizer prediction so that each step adds meaningful insight to realize optimal agricultural yields. The custom CNN model did really well with about 92.88% accuracy when it came to classifying soils. It managed to separate different soil textures and color ranges pretty effectively. These elements work as main signs for soil fertility and the levels of nutrients it holds. In the section on recommending crops, the Random Forest algorithm delivered strong performance. The model achieved 92.4% accuracy on average. It leveraged a variety of inputs towards that, such as Nitrogen shortened by folks to N, Phosphorus quoted as P, and Potassium just referred to by the letter K. The platform also included things like Temperature, Humidity, and Rainfall to complete the picture. With all that fitting together, it was able to identify the best crop choices for specific soil configurations and weather trends. For predicting fertilizer needs, the XGBoost model proved the strongest choice. It achieved 94.7% accuracy. That came from its gradient boosting technique and the regularization features it includes. Those elements make it great for giving precise nutrient advice across various crop types. The project also draws on Explainable AI methods. These cover SHAP for SHapley Additive exPlanations. They include LIME as Local Interpretable Model-Agnostic Explanations. They bring in more clarity and build confidence around the models predictions. These tools graphically emphasize the relative significance of features such as Phosphorus, Potassium, Humidity, and Rainfall in determining the final recommendation, along with pinpointing prominent pixel areas in soil images that influence classification. This ensures that the decision-making process is not a “black box” but a transparent and data-driven approach aligned with real agronomic logic.

The combined results affirm that the system not only provides high accuracy but also explainability, scalability, and adaptability, which makes it extremely suitable for practical agricultural use. With the precision of AI models and the interpretability of XAI combined, the system enables farmers and agronomical researchers to make sustainable, efficient, and informed decisions about crop choice and fertilizer control. Furthermore, incorporating Agentic AI further boosts the autonomy of the system by allowing for adaptive decision-making and ongoing learning from real-time field data. Through dynamic reasoning and goal-directed actions, the agentic layer enables the platform not only to predict but to also optimize agricultural decisions based on changing environmental and soil conditions. Accompanying this intelligence, the interactive web interface offers a user-friendly and accessible medium for researchers and farmers. The home page of the platform shows live weather and environmental information, whereas the analysis and recommendation pages for crops enable users to enter regional and soil image data so that predictions and visual reasoning are immediately returned. Collectively, the agentic framework and web-based interface turn the system into a comprehensive, user-centered ecosystem that aligns the cutting-edge AI capabilities with real-world agricultural functionality.

## Data availability

The datasets used in this study are publicly available and can be accessed at the following sources:- Soil Classification Dataset (Dataset-1): 1,563 labeled soil images covering black, red, clay, and alluvial soil types. Available at: <https://www.kaggle.com/datasets/jayaprakashpondy/soil-image-dataset/data>- Crop Recommendation Dataset (Dataset-2): 2,200 structured instances with features such as N, P, K, pH, temperature, humidity, and rainfall, along with crop labels. Available at: <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>- Fertilizer Recommendation Dataset (Dataset-3): 1,200 samples including soil type, crop type, soil nutrients, and corresponding fertilizer labels. Available at: <https://www.kaggle.com/code/mohitsingh1804/fertilizer-recommendation-system>.

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## Author contributions

N.L.P.S, V.G.S and D.N.S. done the conceptualization of this study, prepared the methodology, development of decision system. V.G.S and D.N.S prepared the figures and tables. N.L.P.S and V.G.S done the performance evaluation and documentation. L.R.P. supervised the research, done the performance evaluation and documentation support. All authors reviewed the manuscript.

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## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

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